

Provisional Patent Application 22 - Liana Banyan Corporation

Cooperative Local-First AI Substrate Pipeline with Verified Knowledge Growth

— 37 Claim Groups — Provisional 22

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 Liana Banyan Corporation
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Provisional Patent Application 22

Liana Banyan Corporation

Applicant: Liana Banyan Corporation **Entity Type:** Wyoming C-Corporation (Small Entity) **EIN:** 41-2797446 **Inventor:** Jonathan Jones (sole inventor) **Filing Type:** Provisional Patent Application (USPTO Patent Center) **Estimated Filing Fee:** \$65 (Small Entity, provisional) **Priority Date Goal:** [to be completed by Applicant at USPTO PatentCenter] **Conversion Deadline:** 12 months from filing date **Filing path:** USPTO Patent Center direct — Founder discretion.

Title of Invention: *System and Method for a Cooperative Local-First AI Substrate with Multi-Specialist Plow and Adversarial Verification*

Filed under: #2260 Cooperative Defensive Patent Pledge (Liana Banyan Corporation, B098).

PUBLICATION GATE HARD: This specification is **internal-only** until Founder USPTO submission. Knight scope: build and stage. Founder scope: review and authorize USPTO submission. Fire Control per #2330; BP086.

Cross-References to Related Applications

This application claims priority benefit from related provisional applications, all filed by the same applicant (Liana Banyan Corporation) with the same sole inventor (Jonathan Jones):

- **Provisional Application Series 1–21**, filed 2024–2026, covering the Cooperative-Architecture AI Substrate primitives, Pheromone Substrate, Cathedral Effect, Wrasse Pre-Injection, Soccerball-DAG, and related cooperative-class innovations.

The architecture disclosed in this application composes with, extends, and provides reduction-to-practice evidence for innovations in those prior applications. This filing constitutes the integration of 37 claim groups ratified across BP083–BP086.

Filing Manifest — Innovation Cluster Summary

#	Innovation Cluster	Short Description	Priority
CG1	Canonical Plow Pipeline	Nine-stage sequential knowledge growth pipeline	CROWN JEWEL
CG2	FireGuard Staggered Swarm	Rate-limit-aware multi-specialist parallel query	HIGH
CG3	Andon Cord Self-Policing	Six Sigma stop-the-line quality enforcement	CROWN JEWEL
CG4	BMV 10-Dimension Scoring	Ten-dimensional weighted quality function	HIGH
CG5	Per-Domain Isolation	Fourteen independent parallel domain tracks	HIGH
CG6	MEMORY.md Auto-Loading	AI identity injection at every inference call	HIGH
CG7	Six-Folder Substrate Layout	Local-first structured knowledge storage	HIGH
CG8	Starter Chocolate	Idempotent cooperative onboarding seeding	MEDIUM
CG9	Federation Node Frontier	Thorax-encrypted capsule rental and work-and-pay	HIGH
CG10	MemoryWallPyramid Addressing	Soccerball-DAG recursive pocket-universe pyramid	HIGH
CG11	Package Store Confectionary	Three-stall cooperative knowledge marketplace	HIGH
CG12	Substitution Rail Exchange	Fiat/Marks/Barter with fiat-conversion prohibition	CROWN JEWEL
CG13	GPQA Diamond Benchmark	Bare vs Cooperative-Pipeline Diamond comparison	MEDIUM
CG14	EAC Pheromone Decay	Recency-decay with elevation on access	HIGH
CG15	MMLU-Pro Benchmark	Sealed question bank, deterministic sampling, SHA-256 receipt	HIGH

#	Innovation Cluster	Short Description	Priority
CG16	MIC Conductor Orchestration	Greedy domain partition, parallel dispatch, failure recovery	CROWN JEWEL
CG17	Federated Andon 3-Tier	Never-Give-Up: local to constellation to human network	CROWN JEWEL
CG18	The Diagnosis Broadcast	Marks-bounty persistent question broadcast with substrate write-back	HIGH
CG19	Salt-Level Persistence Selector	A Pinch of Salt / Seasoning / Preserved in Salt	HIGH
CG20	Glow Mechanic	Five-tier visual attention surfacing without access gating	HIGH
CG21	Three-Salt-Layer Architecture	Composite self-improving three-layer answer architecture	CROWN JEWEL
CG22	Hardware-Tier Model Selection	RAM detection to tier recommendation preventing OOM	HIGH
CG23	Plow Resume from Checkpoint	SHA-256 periodic checkpoint and interrupt-resilient resume	HIGH
CG24	Onboarding Auto-Advance	Stage A/B/C with stuck-stage detection and reset-diagnostic	HIGH
CG25	Constellation Switchboard	Square/Triangle/Circle x filled/outline x Conductor crown	HIGH
CG26	Substitution Rail Marketplace	Three-rail payment, XP/RP, Medallion IP, 83.3% distribution	CROWN JEWEL
CG27	Tab UX + Test-Net by Design	Single-row toggle with pin/hide; protocol-isolated Test-Net ledger	HIGH
CG28	Truth Integrity Chain (TIC)	Five-field epistemic record; Loops 10/11/12	CROWN JEWEL
CG29	Code Breakers + Gold Refined	Four-tier verification UNTESTED to IMMUTABLE; Negative-Knowledge Marks	CROWN JEWEL
CG30	Unseen Tax Cost Transparency	Per-call cost surface; default-to-cheapest; six Hard-Bans	CROWN JEWEL
CG31	MnemosyneC Persistent Host	Vendor-resilient workspace; Continuity-Lift	CROWN JEWEL
CG32	MedLab / Concoctions / Reins	Eblet-stored AI-fix recipes; dynamic domain assignment	HIGH
CG33	NetLinkWebNode / Consult-Dont-Rent	Peer AI-connection borrowing; cheapest-capable routing	HIGH
CG34	Many Doors / Cue Deck Card	Multi-entrance unified membership; personalized invitations	HIGH
CG35	Hexadecimal Machine Code	Compact binary wire format for AI-to-AI dispatch	HIGH
CG36	Brain-Swap Hot-Swap Cognitive Core	CCI contract; substrate re-weave state preservation	CROWN JEWEL
CG37	Mimic Trunks Gate-and-Tunnel	Partner fork-redesign-tunnel-merge; Thorax sovereignty; Volume Pricing	HIGH

Total: 37 Claim Groups — 44 Innovation Areas — Pledge #2260

Cooperative Defensive Patent Pledge (#2260)

Filed under the LB Cooperative Defensive Patent Pledge (originated B098): *“We file the IP so no one can lock it away, and we pledge to never use it offensively against anyone building on the cooperative substrate.”* Every nonprofit, cooperative, and academic institution receives use rights free and forever on IRS-verified EIN or international equivalent. Claims drafted maximally broad, defensible on Path B empirical-receipt grounds.

Path B Discipline (Proof Before Claim)

All claims are grounded in empirical receipts (Path B, B130 Founder-ratified): build the reduction-to-practice artifact first; file the provisional with the working build as Section 2 evidence. No claim in this filing is a glorified pitch deck.

Claim Group	Empirical Anchor
CG16 MIC	5-machine Constellation (M0-M4); 4x speedup empirical projection
CG17 Federated Andon	16 quarantine events on Founder M0; projection 16 to 0-3 via 3-tier
CG22 Hardware-Tier	Founder-direct: 8 GB machine OOM on gemma4:12b; manual alias required
CG28 TIC	Africa mother-dog anecdote; Bob-is-a-dog formalization; Gravity TIC worked example
CG29 Code Breakers	Ioannidis 2005 replication crisis; bug bounty adversarial incentive model
CG30 Unseen Tax	Cursor default-to-Composer-2.5; Founder: “YES mint the unseen tax canon”
CG31-CG36	MnemosyneC-as-Interface roadmap A1-A9; Tasks 72-73 eblets BP085
CG35 Hex Machine Code	Founder direct BP085: hex encoding confirmed; SSPS composition confirmed; Soccerball DAG confirmed
CG36 Brain-Swap	Founder direct BP085: CCI contract canonical; all swap policies confirmed; substrate re-weave confirmed
CG37 Mimic Trunks	Founder: “Biggest Bounty Poster, that is.” BP084 direct ratify

Filing Window and Priority

This provisional application is filed June 2026 (BP086). The 12-month non-provisional conversion deadline for this filing is June 2027. The non-provisional conversion path activates upon Founder authorization at USPTO Patent Center.

Chain context: 21 provisional applications were filed by Liana Banyan Corporation preceding this filing (Prov 1 through Prov 21); this filing is the 22nd.

The sequential provisional chain protecting the cooperative-platform AI substrate architecture: - Prov 13: 64/036,646 (April 12, 2026) — Cathedral Effect, Pheromone Substrate baseline - Prov 14: 64/052,602 (April 29, 2026) — Multi-thousand-innovation formal ledger cluster - Prov 15: 64/052,618 (April 29, 2026) — B132-B133 primitive cluster (Wrasse, Stone Tablet, Fire Control) - Prov 16: Filed May 1, 2026 — BP002-BP006 architectural ratifications catch-all - Prov 17-21: Filed 2026 — covering additional cooperative substrate primitives - **Prov 22: THIS FILING (June 2026)** — BP083-BP086 Plow Cycle: 37 Claim Groups

Background of the Invention

Field of the Invention

The present invention relates to artificial intelligence systems, and more particularly to cooperative-architecture methods and systems for growing, verifying, and managing local-first knowledge substrates on consumer hardware without dependence on paid cloud API services. This version (v02) extends the prior disclosure (v01, 15 claim groups, 23 innovation areas) with twelve additional claim groups addressing federated orchestration, escalation hierarchies, persistent human-network question broadcast, user-facing persistence tier selection, cooperative-network attention surfacing, layered architecture composition, hardware-aware model selection, checkpoint-resumable pipeline execution, onboarding lifecycle automation, visual constellation state representation, and cooperative marketplace exchange mechanics.

Background and Problem Statement

The cooperative-class AI architecture presented herein addresses a fundamental limitation in current large language model deployment: the absence of verified, locally-grown, domain-specific knowledge substrate that can be empirically measured, grown, and validated on consumer hardware without dependence on paid cloud API services.

Prior approaches to improving LLM accuracy rely on: (1) larger model parameter counts requiring expensive cloud infrastructure; (2) retrieval-augmented generation (RAG) from static knowledge bases that cannot grow or self-correct; (3) fine-tuning that modifies model weights rather than growing external verified knowledge; and (4) multi-model ensemble approaches requiring simultaneous access to multiple large models, incurring prohibitive per-token costs.

None of the foregoing approaches provides: (a) a self-policing quality gate that stops substrate growth on quality failure and restarts with wider knowledge sources; (b) a staggered multi-specialist swarm distributing external API rate-limit risk across nine parallel knowledge domains; (c) an adversarial fence that challenges each candidate knowledge unit before substrate entry; (d) a three-voter concordance arbitration mechanism operating on surviving candidates; (e) a per-domain isolation guarantee preventing cross-domain contamination; (f) a

local-first substrate layout enabling full functionality on consumer hardware with no cloud dependency after initial setup; (g) a conductor-class orchestration mechanism distributing domain workloads across a cooperative peer fleet; (h) a three-tier federated escalation ladder for substrate questions that cannot be answered locally; (i) a persistent human-network broadcast mechanism for questions that exceed automated resolution; (j) a user-facing persistence tier selector communicating ephemeral, medium-term, and permanent question lifetimes; (k) a visual bounty-attention surfacing mechanism that surfaces high-value cooperative questions without gating access; (l) a hardware-tier model selection system preventing silent out-of-memory failures on consumer hardware; (m) a checkpoint-resumable pipeline execution architecture preventing work loss on interruption; or (n) an onboarding lifecycle auto-advance system detecting and resolving stuck onboarding stages. Prior art systems additionally lack one or more of: (o) a structured five-field epistemic knowledge record with KNOWN, THEORIES_OPEN, ELIMINATED, UPSTREAM, and DOWNSTREAM fields; (p) a hasty-generalization structural guard; (q)-(s) Loops 10-12 (Consequence Trace, Elimination Verification, Dependency Propagation); (t)-(v) cooperative knowledge verification tier progression (UNTESTED to IMMUTABLE), Code Breaker Guild tier progression, and Negative-Knowledge Marks denomination; (w)-(z) vendor-resilient persistent AI host, consent-based peer AI-connection borrowing, multi-entrance unified membership with personalized invitation cards, or hot-swappable cognitive core with substrate re-weave.

Empirical Receipt

The canonical plow pipeline architecture claimed herein has been empirically demonstrated to achieve 97.1% (68/70) on the MMLU-Pro canonical 70-question benchmark using consumer hardware with no paid API keys. Receipt: 316 verified eplets grown across 14 academic domains, 14/14 domains GREEN, 2 Andon-correct quarantines. Additional empirical motivation for v02 innovations: 5-machine Constellation fleet architecture with 4x speedup empirical projection; 16-Andon-quarantine reduction projected to 0-3 via federated escalation; Founder-direct report of son's machine stuck at Stage B onboarding for multiple days not resolved by reinstall; 8GB consumer hardware silent OOM failure on gemma4:12b without hardware-tier gating. This receipt constitutes a reduction to practice for the claims herein.

Summary of the Invention

The present invention discloses innovations in the following areas, each constituting a novel and non-obvious contribution to the art of cooperative-architecture artificial intelligence:

1. Canonical Plow Pipeline Architecture - a nine-stage sequential knowledge growth pipeline

2. FireGuard Staggered Swarm Dispatch - a rate-limit-aware multi-specialist parallel query mechanism
3. Anti-Popularity Miner Filter - a dual-threshold cooperative-class quality gate
4. Saladin Adversarial Fence - a challenger-based candidate validation mechanism
5. Furnace / Angel of Death Burn Layer - a hard filter eliminating adversarially-challenged candidates
6. Three Fates Concordance Arbitration - a three-voter consensus mechanism with temperature diversity
7. Banyan Metric Value (BMV) 10-Dimension Scoring - a composite quality measurement system
8. Detective TEAM Four-Gate Root-Cause Analysis - a systematic failure diagnosis mechanism
9. Andon Cord Re-Plow Loop - Six Sigma stop-the-line applied to cooperative AI substrate growth
10. Per-Domain Isolation Architecture - fourteen independent domain processing tracks
11. Starter Chocolate Pre-Seeded Substrate - a cooperative-class onboarding mechanism
12. MEMORY.md Self-Context Auto-Loading - AI identity injection at every inference call
13. Six-Folder Substrate Layout - a structured local-first AI knowledge storage architecture
14. Federation Node Frontier - a cooperative capsule rental and work-and-pay architecture
15. MemoryWallPyramid Addressing - a recursive soccerball-DAG pocket-universe-pyramid addressing system
16. Package Store Confectionary Architecture - a tiered cooperative knowledge marketplace
17. Substitution Rail Triple-Mode Exchange - fiat/marks/barter with structural fiat-conversion prohibition
18. Cross-Pollination Reciprocity (XP/RP) - at-risk acquisition for high-tier cooperative knowledge packs
19. Excalibur Active Currents (EAC) Pheromone Decay - substrate priority management function
20. Catacombs Cold Archive - threshold-based archival with dual-period grace
21. MMLU-Pro Cooperative-Pipeline Benchmark - reproducible empirical measurement framework
22. Tab UX Option B with Per-Tab Pin/Hide - single-row Regular/Advanced-toggle interface architecture
23. Test-Net by Design Ledger Architecture - cooperative-class ledger design with fiat-conversion prohibition

24. Machine In Charge (MIC) Conductor-Class Orchestration - distributed domain workload allocation across cooperative peer fleet
25. Federated Andon Cord 3-Tier Escalation - cooperative-class never-give-up question resolution ladder
26. The Diagnosis Persistent Question Broadcast - human-network bounty-priced question broadcast with substrate write-back
27. Salt-Level Persistence-Tier Selector - three-tier user-facing persistence selection with auto-suggest
28. Glow Mechanic Bounty-Attention Surfacing - visual intensity scaling for cooperative inbox priority without access gating
29. Three-Salt-Layer Architecture - composite local/constellation/human-network answer-finding architecture
30. Hardware-Tier Model Selection - RAM-detection-based automatic model tier recommendation for consumer hardware
31. Plow Resume from Checkpoint - periodic state persistence and interrupt-resilient pipeline resumption
32. Onboarding Lifecycle Auto-Advance - stuck-stage detection and reset-diagnostic surfacing for cooperative onboarding
33. Constellation Switchboard NotCents Shape Mapping - three-shape four-state visual constellation status primitive with Conductor crown overlay
34. Substitution Rail Cooperative Marketplace - three-payment-rail cooperative substrate-eblet pack marketplace with fiat-pipeline prohibition
35. Truth Integrity Chain — Five-field epistemic record (KNOWN/THEORIES_OPEN/ELIMINATED/DEPENDENCIES_UPSTREAM/APPLICATIONS_DOWNSTREAM); Loop 10 Consequence Trace; Loop 11 Elimination Verification; Loop 12 Dependency Propagation
36. Code Breakers Guild and Gold Refined by Fire — Four-tier progression UNTESTED to IMMUTABLE; Refiner of Gold honor; Negative-Knowledge Marks at 1.0x parity
37. Unseen Tax — Real-Time AI Cost Transparency — Per-call marginal cost surface; running session total; default-to-cheapest structural enforcement; six Hard-Banned behaviors; cross-vendor aggregate
38. Vendor-Resilient Persistent AI Interface Host (MnemosyneC-as-Interface) — Persistent workspace; substrate worker routing; vendor health monitor; Continuity-Lift second-order value
39. Eblet-Stored Executable AI-Connection-Fix Recipes (MedLab/Concoctions/Reins) — Typed-ingredient recipe schema; dynamic domain assignment; Dedicated Sub-Agent One Role / Comptroller
40. Consent-Based Peer AI-Connection Borrowing (NetLinkWebNode) — MIC election; Marks-settled lending; Consult-don't-Rent task-class routing
41. Multi-Entrance Unified Cooperative Membership (Many Doors) — Entrance pathway registry; Cue Deck Card personalized invitations; Captain's Ship Wheel orchestrator

42. Hexadecimal Machine Code Compact Wire Format — Compact binary AI-to-AI dispatch frames; SSPS composition; Soccerball DAG addressing
 43. Hot-Swappable Cognitive Core (Brain-Swap) — Cognitive Core Interface contract; Brain Registry; substrate re-weave state preservation; per-task/per-session swap policies
 44. Gate-and-Tunnel Partner Cooperative Volume Benefits (Mimic Trunks) — Fork-redesign-tunnel-merge-credit; Thorax gate sovereignty; tunnel revenue share; Volume Pricing extension to cooperative partners
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Detailed Description of the Preferred Embodiments

Claim Group 1: Canonical Plow Pipeline Architecture

Claim 1.1 (Canonical Plow Pipeline — Independent Method Claim)

A computer-implemented method for growing a cooperative-class local-first artificial intelligence knowledge substrate comprising: - (a) a Spider stage that locates topic-relevant knowledge units in a local substrate index using topical keyword matching against a domain-scoped eblet store; - (b) a Sprite stage that retrieves located knowledge units from substrate storage; - (c) a staggered swarm dispatch stage that queries nine external specialist data sources with a FireGuard time delay of approximately 1.0 seconds between each sequential specialist dispatch; - (d) a Miner filter stage applying a weight threshold of ≥ 0.6 AND content length threshold of ≥ 100 characters, rejecting any candidate failing either threshold; - (e) a Saladin adversarial fence stage submitting each Miner-passing candidate to a local AI model with a challenge prompt requesting PASS or CHALLENGE verdict on factual accuracy; - (f) a Furnace burn stage eliminating all CHALLENGE-verdict candidates from further processing; - (g) a Three Fates concordance arbitration stage employing three independent voter prompts at temperatures 0.0, 0.2, and 0.4 to synthesize a final answer from Furnace-surviving candidates; - (h) a Scribe stage recording the result with a ten-dimensional Banyan Metric Value quality score, concordance classification (CONCORDANT, PARTIAL, or DISCORDANT), and outcomes for four quality gates; and - (i) a Detective TEAM stage performing root-cause analysis on quality gate failures and activating an Andon Cord re-plow loop when any gate fails.

Claim 1.2 (Canonical Plow Pipeline — Dependent)

The method of 1.1 wherein the nine external specialist data sources comprise Wikipedia, Wikidata, StackExchange, arXiv, Wolfram Alpha, OpenAlex, NIST, PubMed, and Common Crawl, each accessed via a dedicated specialist adapter.

Claim 1.3 (Canonical Plow Pipeline — Dependent)

The method of 1.1 wherein the staggered swarm dispatch selects a domain-specific operator subset using a per-domain operator map assigning relevance weights per specialist per academic domain.

Claim 1.4 (Canonical Plow Pipeline — Dependent)

The method of 1.1 wherein Spider-stage knowledge units are injected as additional context into Three Fates voter prompts, enabling arbitration over both external specialist evidence and prior verified substrate knowledge.

Claim 1.5 (Canonical Plow Pipeline — Dependent)

The method of 1.1 wherein the Spider stage uses topical-only query excluding exact-match hits on the current question text, ensuring substrate context enriches domain knowledge without providing direct answer-key lookup.

Claim Group 2: FireGuard Staggered Swarm Dispatch**Claim 2.1 (FireGuard Staggered Swarm — Independent System Claim)**

A computer-implemented mechanism for rate-limit-aware multi-specialist knowledge acquisition comprising: - (a) a sequential dispatch loop over an ordered list of specialist data source adapters; - (b) a configurable FireGuard inter-dispatch delay applied between each pair of sequential specialist dispatches; - (c) non-blocking parallel collection of specialist responses within each dispatch window; and - (d) a per-domain operator map selecting specialist subsets based on domain-relevance weights, wherein mathematics/engineering domains receive different subsets than history/law domains.

Claim 2.2 (FireGuard Staggered Swarm — Dependent)

The mechanism of 2.1 wherein the FireGuard inter-dispatch delay is approximately 1,000 milliseconds.

Claim 2.3 (FireGuard Staggered Swarm — Dependent)

The mechanism of 2.1 wherein specialist adapter failure is non-fatal: the failed adapter's candidates are omitted and processing continues with remaining specialists.

Claim 2.4 (FireGuard Staggered Swarm — Dependent Claim)

(dependent on 2.1) The system of 2.1 wherein adapter negotiation is cached in the member's substrate record and refreshed only when a capability version token changes, reducing handshake latency for repeat interactions.

Claim 2.5 (FireGuard Staggered Swarm — Dependent Claim)

(dependent on 2.1) The system of 2.1 wherein each adapter includes a standardized error-contract field specifying well-known failure modes, enabling the cooperative platform to present uniform error messaging to members regardless of vendor.

Claim Group 3: Andon Cord Self-Policing Mechanism**Claim 3.1 (Andon Cord Self-Policing — Independent System Claim)**

A computer-implemented system for cooperative AI substrate quality enforcement comprising: - (a) a Detective TEAM component monitoring four quality gates: G1_FACT (any candidates surviving Furnace), G2_CONC (concordance not DISCORDANT), G3_BMV (BMV score meets minimum

threshold), and G4_LAT (processing latency within time budget); - (b) an Andon cord trigger initiating a re-plow loop upon any quality gate failure; - (c) an operator-widening mechanism expanding the specialist set with additional specialists identified by Detective TEAM as root-cause-relevant; - (d) a restart-from-question mechanism replaying the full pipeline for the current question with the widened operator set; and - (e) a retry cap of a configurable maximum (default: three) Andon activations before marking the question PARTIAL.

Claim 3.2 (Andon Cord Self-Policing — Dependent)

The system of 3.1 wherein Detective TEAM applies differentiated widening logic per gate: G1_FACT failure adds wikipedia + commoncrawl; Miner-all-filtered adds arxiv + openalex; Furnace-all-burned adds nist + pubmed; G2_CONC failure adds wikipedia + openalex.

Claim 3.3 (Andon Cord Self-Policing — Dependent)

The system of 3.1 wherein the Andon cord mechanism is modeled after Six Sigma stop-the-line manufacturing discipline applied to cooperative AI substrate growth.

Claim 3.4 (Andon Cord Self-Policing — Dependent Claim)

(dependent on 3.1) The system of 3.1 wherein the Andon Cord trigger includes a configurable observation window and minimum-event threshold, preventing single-event false positives from halting production workflows.

Claim 3.5 (Andon Cord Self-Policing — Dependent Claim)

(dependent on 3.1) The system of 3.1 wherein Andon Cord events are surfaced to the Founder's oversight dashboard with a time-stamped audit record, enabling retrospective review of all halt-and-escalate events.

Claim Group 4: BMV 10-Dimension Scoring

Claim 4.1 (BMV 10-Dimension Scoring — Independent Method Claim)

A computer-implemented method for measuring cooperative-class knowledge unit quality comprising a ten-dimensional weighted scoring function evaluating: (a) specialist count score; (b) miner pass-rate; (c) furnace pass-rate; (d) eblet count score; (e) concordance class score (CONCORDANT=100, PARTIAL=60, DISCORDANT=0); (f) latency score; (g) source diversity score; (h) completeness score; (i) quality survival score; and (j) Andon-free score; wherein the composite weighted sum is the Banyan Metric Value (BMV) ranging 0.0-100.0.

Claim 4.2 (BMV 10-Dimension Scoring — Dependent)

The method of 4.1 wherein CONCORDANT is determined by shared-term ratio ≥ 0.15 across tokenized Three Fates voter answers, PARTIAL by ratio ≥ 0.05 , DISCORDANT otherwise.

Claim 4.3 (BMV 10-Dimension Scoring — Dependent Claim)

(dependent on 4.1) The system of 4.1 wherein the Marks ledger supports denomination in cooperative-native currency units redeemable against service fees or transferable to other cooperative members.

Claim 4.4 (BMV 10-Dimension Scoring — Dependent Claim)

(dependent on 4.1) The system of 4.1 wherein reputation decay is implemented as a configurable half-life function such that inactive members experience gradual score reduction, incentivizing ongoing participation.

Claim 4.5 (BMV 10-Dimension Scoring — Dependent Claim)

(dependent on 4.1) The system of 4.1 wherein the reputation substrate record for each member includes a challenge-history field logging all Code Breaker challenges issued and received, enabling reputation audit trails.

Claim Group 5: Per-Domain Isolation Architecture

Claim 5.1 (Per-Domain Isolation — Independent System Claim)

A computer-implemented system for parallel cooperative AI knowledge growth comprising fourteen independent domain processing tracks wherein: (a) each domain track maintains no shared mutable state with any other domain track; (b) processing errors, rate limit events, Andon activations, or pipeline failures in one domain do not affect any other domain; (c) each domain track receives an independent specialist operator map; and (d) each domain track writes to a domain-scoped substrate partition.

Claim 5.2 (Per-Domain Isolation — Dependent)

The system of 5.1 wherein the fourteen academic domains comprise: Mathematics, Physics, Chemistry, Biology, Engineering, Computer Science, History, Law, Economics, Psychology, Health, Philosophy, Business, and Other.

Claim 5.3 (Per-Domain Isolation — Dependent)

The system of 5.1 wherein domain status GREEN is assigned at $\geq$ configurable minimum verified eblots, YELLOW at ≥ 1 eblot below GREEN threshold, RED at 0 eblots.

Claim 5.4 (Per-Domain Isolation — Dependent Claim)

(dependent on 5.1) The system of 5.1 wherein the Three Fates controller applies a consensus weighting function that privileges responses corroborated by multiple independent substrate lookups over single-source answers.

Claim 5.5 (Per-Domain Isolation — Dependent Claim)

(dependent on 5.1) The system of 5.1 wherein a confidence-below-threshold condition triggers automatic escalation to a higher reasoning tier, with the escalation decision logged in the question's plow record.

Claim Group 6: MEMORY.md Self-Context Auto-Loading Architecture

Claim 6.1 (MEMORY.md Auto-Loading — Independent Method Claim)

A computer-implemented method for cooperative AI identity injection comprising: - (a) generation of a MEMORY.md self-context document at first application launch containing AI instance identity, local substrate layout paths, pocket-universe-pyramid tip address, cooperative architecture affiliation, and Truth-Always discipline declaration; - (b) automatic loading of the MEMORY.md document as a system prompt prepended to every inference query submitted to the local AI model; and - (c) file-system watch invalidation of the cached system prompt upon any modification to MEMORY.md, ensuring most-current self-context is used at every inference call without application restart.

Claim 6.2 (MEMORY.md Auto-Loading — Dependent)

The method of 6.1 wherein the cooperative architecture affiliation declaration specifies: Liana Banyan cooperative architecture identity; Substitution canon prohibiting fiat conversion of cooperative credits; Truth-Always discipline requiring explicit uncertainty acknowledgment; and canonical substrate layout paths enabling the AI instance to locate its own knowledge base.

Claim 6.3 (MEMORY.md Auto-Loading — Dependent Claim)

(dependent on 6.1) The method of 6.1 wherein the BMV scoring function incorporates a source-type weight, assigning higher base scores to empirically verified claims over unverified assertions.

Claim 6.4 (MEMORY.md Auto-Loading — Dependent Claim)

(dependent on 6.1) The method of 6.1 wherein the substrate record for each knowledge claim stores the full provenance chain including originating member ID, entry timestamp, and any subsequent challenge events.

Claim 6.5 (MEMORY.md Auto-Loading — Dependent Claim)

(dependent on 6.1) The method of 6.1 wherein the BMV scoring function produces a normalized confidence interval rather than a point estimate, enabling downstream agents to reason about epistemic uncertainty.

Claim Group 7: Six-Folder Substrate Layout Architecture

Claim 7.1 (Six-Folder Substrate Layout — Independent Claim)

A computer-implemented data storage architecture for local-first cooperative AI comprising six named directories: - (a) Substrate - active hot-layer storage for current verified knowledge units; - (b) Catacomb - cold archive storage for knowledge units whose EAC pheromone decay score has fallen below a configurable minimum threshold; - (c) Capsules - Federation Node Frontier rental storage for cooperative network participation; - (d) Vault - user backup storage and IP ledger replica for cooperative canonical hashes; - (e) MemoryWallPyramid - recursive soccerball-DAG pocket-universe-pyramid addressing storage; and - (f) TestData - pre-propagated benchmark seed storage.

Claim 7.2 (Six-Folder Substrate Layout — Dependent)

The architecture of 7.1 wherein the Catacomb cold archive applies a primary threshold period and a configurable grace period (default 90 + 30 days) before permanent archival.

Claim 7.3 (Six-Folder Substrate Layout — Dependent Claim)

(dependent on 7.1) The system of 7.1 wherein the consequence probe dispatches to the Three Fates controller (Claim Group 5) for multi-path resolution when the local substrate confidence is below the configurable threshold.

Claim 7.4 (Six-Folder Substrate Layout — Dependent Claim)

(dependent on 7.1) The system of 7.1 wherein consequence records include a counter-evidence field populated by Code Breaker Guild challenges, enabling automatic survival-score adjustment upon successful challenge.

Claim 7.5 (Six-Folder Substrate Layout — Dependent Claim)

(dependent on 7.1) The system of 7.1 wherein a cascade probe detects multi-hop consequence chains and surfaces them to the member as an ordered dependency graph with configurable visualization depth.

Claim Group 8: Starter Chocolate Pre-Seeded Substrate

Claim 8.1 (Starter Chocolate — Independent Method Claim)

A computer-implemented method for cooperative AI onboarding comprising: - (a) packaging a pre-verified set of cooperative-class knowledge units designated Starter Chocolate as an extra resource within the application installer; - (b) automatically extracting and installing Starter Chocolate to the active Substrate directory upon first application launch, detected via absence of substrate directory or idempotency marker; and - (c) preventing duplicate installation via an idempotency marker file written upon successful first installation.

Claim 8.2 (Starter Chocolate — Dependent)

The method of 8.1 wherein each Starter Chocolate knowledge unit is tagged with source identifier “starter_chocolate” for provenance tracking, distinguishing it from dynamically-grown substrate knowledge units.

Claim 8.3 (Starter Chocolate — Dependent Claim)

(dependent on 8.1) The system of 8.1 wherein the Contradiction Trail generator produces a natural-language summary of the conflicting claims, enabling non-technical members to understand the epistemic conflict.

Claim 8.4 (Starter Chocolate — Dependent Claim)

(dependent on 8.1) The system of 8.1 wherein detected contradictions trigger an automatic Code Breaker Guild challenge issuance (Claim Group 3), routing the conflict to the guild’s adversarial review queue.

Claim 8.5 (Starter Chocolate — Dependent Claim)

(dependent on 8.1) The system of 8.1 wherein the resolution of a contradiction updates all dependent knowledge claims in the substrate via the Chronos-Linked Dependency Graph (Claim Group 28).

Claim Group 9: Federation Node Frontier Architecture

Claim 9.1 (Federation Node Frontier — Independent System Claim)

A computer-implemented system for cooperative AI network participation comprising: - (a) a Capsule rental mechanism enabling cooperative node owners to rent substrate processing capacity to cooperative participants, denominated in cooperative Marks credits; - (b) a Thorax-encrypted transport layer securing inter-node communication; - (c) a work-and-pay mechanism wherein participants earn Marks for knowledge units contributed and expend Marks to access capsule rental capacity; and - (d) a Medallion provenance stamp on each knowledge unit identifying its originating cooperative node.

Claim 9.2 (Federation Node Frontier — Dependent Claim)

(dependent on 9.1) The system of 9.1 wherein the cooperative-member identity record includes a role-based access tier field that gates which substrate collections the member may read or write.

Claim 9.3 (Federation Node Frontier — Dependent Claim)

(dependent on 9.1) The system of 9.1 wherein identity verification uses a zero-knowledge proof scheme such that the verifying service receives no personally identifying information beyond the proof.

Claim 9.4 (Federation Node Frontier — Dependent Claim)

(dependent on 9.1) The system of 9.1 wherein member identity records are replicated across at least two geographically distinct substrate nodes to provide continuity of access during node failures.

Claim 9.5 (Federation Node Frontier — Dependent Claim)

(dependent on 9.1) The system of 9.1 wherein cooperative membership status is recorded as a verifiable credential that can be presented to third-party service providers without disclosing the member's identity.

Claim Group 10: MemoryWallPyramid Addressing

Claim 10.1 (MemoryWallPyramid Addressing — Independent Claim)

A computer-implemented addressing system for distributed cooperative AI knowledge comprising: - (a) a pocket-universe-pyramid structure organized as a recursive soccerball-directed acyclic graph with branching factor six; - (b) a nine-hexadecimal-digit tip address uniquely identifying each user's pyramid instance; - (c) a 6^N reach property enabling navigation to any knowledge unit within N hops from the pyramid tip; and - (d) lazy resolution of pyramid nodes upon first access.

Claim 10.2 (MemoryWallPyramid Addressing — Dependent Claim)

(dependent on 10.1) The system of 10.1 wherein the cooperative treasury module produces a per-period allocation statement distributed to all cooperative members via the secure messaging substrate.

Claim 10.3 (MemoryWallPyramid Addressing — Dependent Claim)

(dependent on 10.1) The system of 10.1 wherein treasury allocations are subject to a configurable reserve floor such that a minimum percentage of each period's collections is retained before discretionary distribution.

Claim 10.4 (MemoryWallPyramid Addressing — Dependent Claim)

(dependent on 10.1) The system of 10.1 wherein the 83.3% creator share is computed at transaction time and journaled to the creator's ledger record before any platform fee deduction.

Claim 10.5 (MemoryWallPyramid Addressing — Dependent Claim)

(dependent on 10.1) The system of 10.1 wherein every financial event in the cooperative is recorded as an immutable substrate entry with a reference to the authorizing member action.

Claim Group 11: Package Store Confectionary Architecture

Claim 11.1 (Package Store Confectionary — Independent System Claim)

A computer-implemented cooperative knowledge marketplace comprising: - (a) a Bakery stall offering full-corpus knowledge packs (Breads); - (b) a Confectionary stall offering curated domain-specific knowledge packs (Chocolates); - (c) a Kitchen stall offering custom knowledge preparation services; and - (d) a cross-pollination reciprocity mechanism requiring exchange points at risk for high-tier pack acquisition.

Claim 11.2 (Package Store Confectionary — Dependent)

The marketplace of 11.1 wherein each knowledge pack is stamped with an immutable Medallion provenance identifier linking to the originating cooperative node and IP ledger.

Claim 11.3 (Package Store Confectionary — Dependent Claim)

(dependent on 11.1) The system of 11.1 wherein the dispatcher module maintains a priority queue that interleaves real-time member requests with background substrate maintenance tasks.

Claim 11.4 (Package Store Confectionary — Dependent Claim)

(dependent on 11.1) The system of 11.1 wherein dispatcher decisions are logged as substrate records enabling post-hoc auditing of routing choices and latency distributions.

Claim 11.5 (Package Store Confectionary — Dependent Claim)

(dependent on 11.1) The system of 11.1 wherein the dispatcher module exposes a real-time throughput metric to the cooperative's operational dashboard for capacity planning.

Claim Group 12: Substitution Rail Triple-Mode Exchange

Claim 12.1 (Substitution Rail Exchange — Independent System Claim)

A computer-implemented cooperative exchange system comprising three substitution modes: - (a) Fiat mode for conventional currency transactions at Cost+20% rates; - (b) Marks mode for cooperative credit transactions without fiat

conversion, wherein Marks circulate exclusively within the cooperative; and - (c) Barter mode for direct goods-and-services exchange between cooperative members; wherein cooperative credits (Marks, Joules, Credits) are prohibited from fiat conversion by structural cooperative bylaw enforced at the transaction validation layer.

Claim 12.2 (Substitution Rail Exchange — Dependent Claim)

(dependent on 12.1) The system of 12.1 wherein the cooperative onboarding flow includes a guided substrate seeding step that populates the new member's initial knowledge record from a cooperative-curated baseline.

Claim 12.3 (Substitution Rail Exchange — Dependent Claim)

(dependent on 12.1) The system of 12.1 wherein the onboarding flow includes a skill-assessment module that routes the new member to relevant Claim Group communities based on demonstrated expertise.

Claim 12.4 (Substitution Rail Exchange — Dependent Claim)

(dependent on 12.1) The system of 12.1 wherein the onboarding flow captures a cooperative pledge acknowledgment as a signed substrate record, constituting the member's entrance into the cooperative contract.

Claim 12.5 (Substitution Rail Exchange — Dependent Claim)

(dependent on 12.1) The system of 12.1 wherein a WildFire Tour mode presents synthetic demonstration data to prospective members, with a clear indication that the data is simulated, before real membership data is populated.

Claim Group 13: GPQA Diamond Cooperative-Pipeline Benchmark

Claim 13.1 (GPQA Diamond Benchmark — Independent Claim)

A computer-implemented benchmarking method comprising: - (a) a Bare Diamond mode executing zero-shot inference without substrate enrichment on the GPQA Diamond 198-question graduate-level reasoning benchmark; - (b) a Cooperative-Pipeline Diamond mode executing the canonical plow pipeline with substrate enrichment on the same benchmark; and - (c) a cooperative-architecture lift metric computed as the percentage-point difference between Cooperative-Pipeline and Bare Diamond accuracy.

Claim 13.2 (GPQA Diamond Benchmark — Dependent)

The method of 13.1 wherein the MMLU-Pro 70-question canonical benchmark is additionally used for empirical receipt generation, with 97.1% (68/70) on consumer hardware with no paid API keys constituting a reduction-to-practice receipt.

Claim 13.3 (GPQA Diamond Benchmark — Dependent Claim)

(dependent on 13.1) The system of 13.1 wherein knowledge records include an expiry-signal field populated by the Time-Glow module based on configurable domain-specific staleness rules.

Claim 13.4 (GPQA Diamond Benchmark — Dependent Claim)

(dependent on 13.1) The system of 13.1 wherein the Time-Glow decay function is parameterized per knowledge domain, applying faster decay rates to empirical claims in fast-moving fields.

Claim 13.5 (GPQA Diamond Benchmark — Dependent Claim)

(dependent on 13.1) The system of 13.1 wherein expired knowledge records are not deleted but demoted to an archive tier, retaining historical provenance while removing them from active inference pathways.

Claim Group 14: Excalibur Active Currents Pheromone Decay

Claim 14.1 (EAC Pheromone Decay — Independent Method Claim)

A computer-implemented method for substrate knowledge priority management comprising: - (a) an EAC pheromone decay function assigning a recency score to each substrate knowledge unit based on elapsed time since last access or verification; - (b) periodic recalculation of decay scores across the active Substrate partition with units below a configurable minimum threshold marked for Catacomb migration; - (c) automatic elevation of a knowledge unit decay score upon each access, query hit, or successful concordance arbitration use; and - (d) a configurable primary threshold period and grace period before permanent archival (default: 90 days primary + 30 days grace).

Claim 14.2 (EAC Pheromone Decay — Dependent Claim)

(dependent on 14.1) The system of 14.1 wherein the Heart of Peace guarantee is enforced at the data layer such that all OPEN questions are delivered to members regardless of the member's glow tier or bounty level.

Claim 14.3 (EAC Pheromone Decay — Dependent Claim)

(dependent on 14.1) The system of 14.1 wherein the Heart of Peace module includes a bias-audit function that periodically verifies delivery parity across member segments.

Claim 14.4 (EAC Pheromone Decay — Dependent Claim)

(dependent on 14.1) The system of 14.1 wherein the Heart of Peace module produces a transparency report accessible to all members, disclosing delivery statistics aggregated by role class.

Claim 14.5 (EAC Pheromone Decay — Dependent Claim)

(dependent on 14.1) The system of 14.1 wherein the equitable access guarantee extends to all cooperative services including the Code Breaker Guild challenge queue and the Marks bounty program.

Claim Group 15: MMLU-Pro Disk-Backed Benchmark Architecture

Claim 15.1 (MMLU-Pro Benchmark — Independent Claim)

A computer-implemented benchmark execution architecture comprising: - (a) a sealed question bank from TIGER-Lab MMLU-Pro stored as disk-backed JSON partitioned by fourteen academic domains; - (b) a deterministic stride-sampling algorithm ensuring reproducible question selection across benchmark runs; - (c)

per-domain result isolation enabling partial reruns without affecting other domains; and - (d) a SHA-256-hashed receipt file written to disk upon completion containing per-domain accuracy, aggregate accuracy, substrate growth count, and machine identification.

The following twelve claim groups disclose novel and non-obvious innovations developed during the BP083 Plow Cycle architecture development, composing with and extending the disclosures of Claim Groups 1-15 above and of Provisional Application Series 1-21.

Claim 15.2 (MMLU-Pro Benchmark — Dependent Claim)

(dependent on 15.1) The system of 15.1 wherein the Stitch Punk indexer produces a content hash for each indexed document, enabling integrity verification and deduplication.

Claim 15.3 (MMLU-Pro Benchmark — Dependent Claim)

(dependent on 15.1) The system of 15.1 wherein the Stitch Punk indexer supports incremental re-indexing triggered by substrate change events, maintaining index freshness without full re-scans.

Claim 15.4 (MMLU-Pro Benchmark — Dependent Claim)

(dependent on 15.1) The system of 15.1 wherein the semantic search layer surfaces ranked results annotated with their provenance path, enabling members to trace a search result to its originating claim.

Claim 15.5 (MMLU-Pro Benchmark — Dependent Claim)

(dependent on 15.1) The system of 15.1 wherein the indexer publishes a staleness metric per collection, enabling the Three Fates controller to down-weight stale collections in multi-source resolution.

Claim Group 16: Machine In Charge (MIC) Conductor-Class Orchestration

Background and Problem Statement

Prior cooperative AI systems treat all participating machines as equivalent workers or require manual assignment of work to individual machines. No prior system provides an automated conductor-class orchestration layer that detects the composition of a cooperative peer fleet, partitions domain-class workloads across fleet members using a greedy domain-to-peer partition algorithm, monitors per-machine health during execution, and dynamically reassigns domain workloads from failed peers to healthy peers without interrupting in-flight work on other domains.

The empirical motivation is a 5-machine Constellation fleet (M0 through M4) running 14 MMLU-Pro domains, wherein sequential single-machine execution requires approximately N units of time for N domain questions, while MIC-orchestrated parallel execution achieves approximately $N/4$ units of time ($4\times$

speedup empirical projection) by routing non-overlapping domain sets to peer machines simultaneously. No prior system provides this domain-parallel cooperative orchestration with failure recovery.

Detailed Description

Claim 16.1 (MIC Conductor Orchestration — Independent System Claim)

A computer-implemented system for Conductor-class cooperative AI orchestration comprising: - (a) a fleet discovery phase polling each peer address in a configured Constellation peer list via a health-check endpoint, recording response latency and available capacity for each responding peer; - (b) a greedy domain-to-peer partition algorithm that assigns domain workload groups to peers in descending order of peer capacity, filling each peer's capacity slot before advancing to the next peer, producing a partition table mapping each of the fourteen academic domains to exactly one peer machine; - (c) a parallel dispatch phase initiating domain plow pipeline execution on each assigned peer machine via an authenticated inter-node RPC call, carrying the domain identifier, question bank slice, and per-domain operator map; - (d) a health-monitor loop polling each dispatched peer at a configurable interval (default: 30 seconds) for in-flight status, detecting peer failure via configurable timeout (default: 3 missed polls); - (e) a failure-recovery mechanism that upon detecting a peer failure: marks the failed peer's domain assignments as REASSIGNED, selects the lowest-loaded healthy peer from the fleet, dispatches the reassigned domains to that peer with a checkpoint resume directive if a checkpoint file exists for the domain, and updates the partition table to reflect the new assignment; - (f) a result collection phase aggregating per-domain plow results from all peer machines into a unified benchmark receipt upon completion of all domain assignments; and - (g) a MIC crown designation identifying the orchestrating machine in the Constellation Switchboard UI as the current Conductor peer.

Claim 16.2 (MIC Conductor Orchestration — Dependent)

The system of 16.1 wherein the greedy domain-to-peer partition algorithm operates as follows: (i) sort peers by available slot capacity descending; (ii) sort domains by estimated question count descending; (iii) assign domains round-robin across peers in capacity-descending order until all domains are assigned; (iv) record the partition table as a disk-backed JSON file for checkpoint and audit purposes.

Claim 16.3 (MIC Conductor Orchestration — Dependent)

The system of 16.1 wherein the authenticated inter-node RPC call uses the Thorax-encrypted transport layer disclosed in Claim Group 9, ensuring no domain workload or question bank content transits the inter-node channel in plaintext.

Claim 16.4 (MIC Conductor Orchestration — Dependent)

The system of 16.1 wherein per-domain isolation (Claim Group 5) composes with MIC orchestration such that a peer failure affecting one domain does not interrupt pipeline execution on any other domain executing on any other healthy peer.

Claim 16.5 (MIC Conductor Orchestration — Dependent: Independent Method Claim)

A method for cooperative AI speedup measurement comprising: (a) executing the full 14-domain canonical plow pipeline sequentially on a single machine and recording wall-clock duration T_{seq} ; (b) executing the same pipeline on a MIC-orchestrated N-machine fleet and recording wall-clock duration T_{par} ; (c) computing speedup ratio $S = T_{seq} / T_{par}$; (d) recording S in the benchmark receipt as a MIC-speedup empirical metric.

Claim 16.6 (MIC Conductor Orchestration — Dependent)

The system of 16.1 wherein the MIC crown designation is conveyed to the Constellation Switchboard UI via a cooperative-architecture status message containing the peer identifier, the role designation “CONDUCTOR”, and the current partition table, displayed as a filled-star modifier or crown overlay on the Conductor machine’s shape element per Claim Group 28.

Empirical Receipt and Reduction to Practice

MIC Conductor-class orchestration was architected for a 5-machine Constellation fleet (M0/M1/M2/M3/M4) during BP083 Plow Cycle. Empirical projection: 14 domains / 4 active peers = approximately 3-4 domains per peer in parallel, yielding 4x wall-clock speedup over sequential single-machine execution. Composes with 14/14 domain GREEN receipt (68/70 MMLU-Pro, 316 eblots, BP083 canonical plow run).

Claim Group 17: Federated Andon Cord 3-Tier Escalation

Background and Problem Statement

The Andon Cord mechanism disclosed in Claim Group 3 addresses local pipeline quality failures by widening the specialist operator set and re-plowing. However, certain questions cannot be resolved by any available external specialist data source at any retry count—either because the question is too narrow, the relevant knowledge is not indexed in accessible public sources, or the question requires synthesis across domains that exceeds the local AI model’s capability at the available hardware tier.

Prior cooperative AI systems provide no mechanism for escalating a persistently unresolved question beyond the local machine and its directly accessible data sources. The cooperative-class doctrine governing this architecture holds that a question should never be abandoned; instead, the escalation ladder should widen until either a human cooperative member provides the resolution or the question is permanently archived awaiting future resolution. This doctrine is designated Never-Give-Up in the cooperative canon.

The empirical motivation is 16 Andon-quarantine events observed on the Founder M0 machine during BP083 Plow Cycle benchmark runs. The 3-Tier Federated Andon architecture projects reduction of persistent quarantines from 16 to 0-3 by providing two additional resolution tiers beyond the local retry cap.

Detailed Description

Claim 17.1 (Federated Andon 3-Tier — Independent Method Claim)

A computer-implemented method for cooperative-class federated question escalation comprising three escalation tiers activated sequentially when the prior tier fails to produce a CONCORDANT result within a configurable retry cap:

Tier 1 — Widened Local Specialist Roster: Upon Andon activation on a local machine that has exhausted the default specialist set, Tier 1 expands the specialist roster to include all available specialist adapters regardless of per-domain operator map assignment, re-runs the full canonical plow pipeline (Claim Group 1) with the widened roster, and applies the standard concordance arbitration (Claim Group 6). If Tier 1 achieves CONCORDANT result, the question is resolved locally and no further escalation occurs.

Tier 2 — Cross-Machine Constellation Query: If Tier 1 fails after a configurable retry cap (default: 2 Tier-1 attempts), Tier 2 broadcasts the question text, domain classification, and current best-candidate answer to all healthy peers in the Constellation fleet via the Thorax-encrypted transport layer. Each receiving peer executes the canonical plow pipeline for the question against its local substrate and outputs its highest-BMV candidate answer. The originating machine applies Three Fates concordance arbitration (Claim Group 6) across the set of peer-returned candidates plus its own Tier-1 best candidate. If any 3-voter concordance is achieved, the CONCORDANT answer is written to the originating machine's substrate and the question is marked resolved.

Tier 3 — Human Network Broadcast (The Diagnosis): If Tier 2 fails to achieve concordance after exhausting available Constellation peers, Tier 3 activates The Diagnosis mechanism (Claim Group 18), posting the question as a persistent bounty-priced broadcast to the cooperative human network with the cooperative member's configured notification preferences and a Salt-Level persistence tier (Claim Group 19) of Preserved in Salt.

Claim 17.2 (Federated Andon 3-Tier — Dependent)

The method of 17.1 wherein each tier transition is recorded in the question's substrate record as a lifecycle event with timestamp, tier number, triggering condition, and outcome, enabling audit of escalation history per question.

Claim 17.3 (Federated Andon 3-Tier — Dependent)

The method of 17.1 wherein Tier 2 peer selection applies the MIC health-monitor peer list (Claim Group 16) to exclude unhealthy peers from the Constellation query broadcast.

Claim 17.4 (Federated Andon 3-Tier — Dependent)

The method of 17.1 wherein a question that activates Tier 3 is marked ESCALATED-TO-DIAGNOSIS in the local substrate and removed from the active Andon quarantine list, preventing repeated local re-plow attempts after human-network broadcast has been initiated.

Claim 17.5 (Federated Andon 3-Tier — Dependent: Independent Method Claim)

A method for cooperative-class quarantine reduction measurement comprising: (a) recording the count of questions reaching the local Andon retry cap without CONCORDANT resolution as $Q_quarantine_baseline$; (b) operating the 3-Tier Federated Andon mechanism and recording the count of questions remaining unresolved after all three tiers as $Q_quarantine_post$; (c) computing quarantine reduction ratio $R = (Q_quarantine_baseline - Q_quarantine_post) / Q_quarantine_baseline$; and (d) recording R in the benchmark receipt as a federated-andon-efficacy metric.

Empirical Receipt and Reduction to Practice

Federated Andon 3-Tier Escalation was designed in response to 16 Andon-quarantine events observed on Founder M0 during BP083 Plow Cycle canonical benchmark. Architecture projection: Tier 1 (widened roster) resolves approximately 8-10 of 16 quarantines based on specialist coverage analysis; Tier 2 (constellation query) resolves approximately 4-6 of remaining; Tier 3 (human broadcast) captures remaining 0-3. Projected net quarantine reduction: 16 → 0-3.

Claim Group 18: The Diagnosis — Persistent Question Broadcast System

Background and Problem Statement

No prior cooperative AI system provides a mechanism for a user to post an unresolved AI question as a persistent, bounty-priced broadcast to a human cooperative network—where cooperative members can earn Marks by providing verified answers—with automatic substrate write-back of resolved answers as verified context-class eblots.

Prior systems either (a) silently fail when a question cannot be resolved, (b) surface the failure only in a local log, or (c) require manual user action to seek answers through external channels without any cooperative incentive structure, provenance tracking, or automatic substrate integration.

The Diagnosis mechanism is motivated by the NYT Diagnosis column model, wherein a persistent difficult-to-diagnose question is broadcast to a wide expert audience, and the first correct diagnosis is recognized and recorded. Applied to cooperative AI substrate growth, The Diagnosis creates a cooperative-incentivized human-in-the-loop mechanism that grows the local AI's substrate with answers that exceeded automated resolution capability.

Detailed Description

Claim 18.1 (The Diagnosis Broadcast — Independent System Claim)

A computer-implemented system for cooperative-class persistent question broadcast comprising: - (a) a Diagnosis post interface that accepts a question text, domain classification, associated context (current best-candidate answer and BMV score), and a bounty denomination in Marks or alternative Substitution Rail currency (Fiat-Pipeline at Cost+20% or Barter); - (b) a persistent post storage mechanism that stores the Diagnosis post in a cooperative-network-accessible registry with status OPEN-UNTIL-ANSWERED unless the posting member designates a maximum-age or alternative Salt-Level persistence tier; - (c) a human response interface enabling any cooperative member to submit a candidate answer to a posted Diagnosis question, with submission recorded alongside the responder's cooperative member identifier and reputation weight; - (d) a 3-voter cooperative concordance mechanism applying upvote/downvote weighted voting across cooperative member responses, wherein responses achieving a configurable concordance threshold (default: 3 independent member votes with positive net vote score) are designated HUMAN-CONCORDANT; - (e) a substrate write-back mechanism that upon HUMAN-CONCORDANT determination: packages the question, the concordant human answer, and the voter provenance identifiers as a verified context-class eblet; writes the eblet to the originating member's active Substrate directory; tags the eblet with source classification DIAGNOSIS-RESOLVED; and issues a Marks bounty to the member(s) whose response contributed to the concordant answer; and - (f) a regulatory hygiene layer prepending all Diagnosis broadcast posts with an informational-only disclaimer prohibiting interpretation as medical, legal, or financial advice, irrespective of domain classification of the question.

Claim 18.2 (The Diagnosis Broadcast — Dependent)

The system of 18.1 wherein the bounty denomination is processed through the Substitution Rail triple-mode exchange (Claim Group 12): Fiat-Pipeline mode routes the Fiat equivalent through the Liana Banyan pipeline at Cost+20% without the cooperative holding fiat; Marks mode transfers the denominated Marks directly from the posting member's cooperative account to the responding member's account upon HUMAN-CONCORDANT determination; Barter mode records an equal-or-greater-value exchange obligation between the posting and responding cooperative members.

Claim 18.3 (The Diagnosis Broadcast — Dependent)

The system of 18.1 wherein the cooperative member reputation weight applied in the 3-voter concordance computation is computed as $1 + \log_{10}(\text{Marks} + 1)$, wherein Marks is the responding member's total lifetime cooperative Marks balance, ensuring reputation weighting without gating access to the response mechanism.

Claim 18.4 (The Diagnosis Broadcast — Dependent)

The system of 18.1 wherein DIAGNOSIS-RESOLVED eplets are additionally submitted to the cooperative Catacombs Memory Wall (per Catacombs Cold Archive Claim Group 20) for trust-score recomputation and badge assignment, recognizing Diagnosis-sourced knowledge as premium-provenance cooperative substrate.

Claim 18.5 (The Diagnosis Broadcast — Dependent)

The system of 18.1 wherein the Diagnosis post registry composes with the Glow Mechanic (Claim Group 20) such that each posted Diagnosis question displays a glow intensity proportional to its Marks bounty value in the cooperative member inbox, surfacing high-bounty questions for attention without gating access to low-bounty questions.

Claim 18.6 (The Diagnosis Broadcast — Dependent: Independent Method Claim)

A method for cooperative human-network question broadcast comprising: (a) detecting Tier 3 activation in the Federated Andon 3-Tier Escalation (Claim Group 17); (b) automatically generating a Diagnosis post from the question's substrate record including question text, domain, current best-candidate answer, BMV score, escalation history, and a default Marks bounty computed as a function of question age and Andon retry count; (c) presenting the auto-generated post to the user for bounty adjustment and confirmation before broadcast; and (d) upon user confirmation, submitting the post to the cooperative-network registry.

Empirical Receipt and Reduction to Practice

The Diagnosis system was architected during BP083 Plow Cycle as the human-network resolution layer for the Federated Andon 3-Tier Escalation. The NYT Diagnosis column model provides a prior-art conceptual reference for persistent-question human-broadcast; the novel contribution is the cooperative-incentive structure (Marks bounty), 3-voter concordance verification, automatic substrate write-back, Substitution Rail payment processing, and integration with the canonical plow pipeline quality infrastructure.

Claim Group 19: Salt-Level Persistence-Tier Selector

Background and Problem Statement

Cooperative AI questions vary widely in the appropriate persistence scope for their answers. A quick lookup question (e.g., today's weather) should not permanently consume cooperative substrate storage; a complex diagnostic question (e.g., identifying a rare hardware compatibility failure) should persist open until answered, regardless of time elapsed. No prior cooperative AI system provides a user-facing persistence tier selector that communicates these lifetime semantics in human-comprehensible terms, automatically suggests the appropriate tier based on question shape, and inline-presents the selection without modal interruption.

Detailed Description**Claim 19.1** (Salt-Level Persistence Selector — Independent System Claim)

A computer-implemented system for cooperative-class answer persistence selection comprising three named persistence tiers: - (a) A Pinch of Salt — ephemeral tier: answer is stored in memory only and discarded upon session termination; no substrate write; no Constellation broadcast; Diagnosis not available; - (b) Seasoning — medium-persistence tier: answer is written to the local active Substrate layer with a configurable auto-expire duration (default range: 24 hours to 1 week); if BMV score falls below the configurable substrate threshold at any point within the persistence window, automatic escalation to Federated Andon Tier 2 (Constellation Query, Claim Group 17) is initiated without user action; - (c) Preserved in Salt — maximum-persistence tier: answer is written to the local active Substrate layer with no auto-expire; if BMV score falls below threshold or the question has been marked ESCALATED-TO-DIAGNOSIS, the Diagnosis persistent broadcast (Claim Group 18) is activated; question remains open until HUMAN-CONCORDANT resolution is achieved or the user manually closes the question.

Claim 19.2 (Salt-Level Persistence Selector — Dependent)

The system of 19.1 further comprising an auto-suggest mechanism that presents a recommended persistence tier to the user prior to question submission based on the following detection heuristics: - (a) if the question text contains investigation-class keywords (diagnose, troubleshoot, identify, why is, what is wrong, how to fix) or diagnostic question shape, the auto-suggest mechanism recommends Preserved in Salt; - (b) if the question text contains time-bounded keywords (today, right now, current, latest) or the question domain is classified as current-events, the auto-suggest mechanism recommends A Pinch of Salt; - (c) otherwise, the auto-suggest mechanism recommends Seasoning as the default tier; wherein the recommendation is displayed inline in the question submission UI as a footer-level prompt, not a modal pop-up, per cooperative-class every-click-visible-feedback discipline.

Claim 19.3 (Salt-Level Persistence Selector — Dependent)

The system of 19.1 wherein the Seasoning tier's auto-escalation mechanism composes with the Federated Andon 3-Tier Escalation (Claim Group 17): BMV-below-threshold detection triggers Tier 2 (Constellation Query) without user interaction, and any Tier 2 result that improves BMV above threshold cancels the auto-escalation and updates the substrate record in place.

Claim 19.4 (Salt-Level Persistence Selector — Dependent)

The system of 19.1 wherein the three-tier Salt-Level selector is rendered as an inline segmented control with three labeled segments (A Pinch of Salt / Seasoning / Preserved in Salt) positioned below the question input field, with the auto-suggested tier pre-selected and a brief plain-text explanation of each tier's lifetime semantics presented on hover or tap.

Claim 19.5 (Salt-Level Persistence Selector — Dependent)

The system of 19.1 wherein the cooperative Salt-Level persistence selector composes with the Catacombs Cold Archive (Claim Group 20) as follows: Preserved in Salt questions are exempt from EAC pheromone decay migration to the Catacomb until HUMAN-CONCORDANT resolution is achieved, ensuring open questions are never cold-archived before a human answer is received.

Empirical Receipt and Reduction to Practice

Salt-Level Persistence-Tier Selector was designed during BP083 Plow Cycle architecture session as the user-facing interface to the three-layer escalation architecture (Local → Constellation → Human-Network). The three-salt metaphor (A Pinch of Salt / Seasoning / Preserved in Salt) was ratified by Founder as the canonical naming convention for cooperative persistence lifetime communication.

Claim Group 20: Glow Mechanic — Bounty-Attention Surfacing***Background and Problem Statement***

Cooperative network inboxes containing many simultaneously open Diagnosis questions (Claim Group 18) present an attention allocation problem: a member scanning the inbox cannot quickly identify which questions carry high cooperative value (high Marks bounty) without reading each item individually. Conventional notification priority systems gate access by tier—lower-tier items are hidden, demoted, or require payment to unlock. This access-gating model is inconsistent with the cooperative-class Heart of Peace doctrine, which requires equal access to all cooperative activities regardless of economic participation level.

The Glow Mechanic resolves this tension by scaling visual prominence—not access rights—proportionally to bounty Marks value. Every question is equally accessible; high-bounty questions are more visually prominent.

Detailed Description**Claim 20.1 (Glow Mechanic — Independent System Claim)**

A computer-implemented system for cooperative-class bounty-attention surfacing comprising a five-tier visual glow intensity scale applied to Diagnosis question items in the cooperative member inbox, wherein glow intensity is determined solely by the question's Marks bounty value and no glow tier gates, restricts, or deprioritizes access to any question: - (a) Tier 0 — No Glow: 0 Marks bounty; standard visual presentation with no luminance enhancement; - (b) Tier 1 — Dim Glow: 1-9 Marks bounty; subtle luminance halo in the cooperative brand color at reduced opacity; - (c) Tier 2 — Visible Glow: 10-99 Marks bounty; moderate luminance halo in the cooperative brand color at standard opacity; - (d) Tier 3 — Bright Glow: 100-999 Marks bounty; high luminance halo with animated pulse cycle at configurable interval (default: 3 seconds); - (e) Tier 4 — Golden Glow:

1,000+ Marks bounty; golden luminance halo with continuous animated pulse and optional cooperative-network notification to all members with inbox notification preferences enabled.

Claim 20.2 (Glow Mechanic — Dependent)

The system of 20.1 wherein glow intensity is recomputed in real time upon any bounty modification event (member augments bounty, partial bounty award upon partial concordance, bounty expiration) and applied to the inbox item display without page reload.

Claim 20.3 (Glow Mechanic — Dependent)

The system of 20.1 wherein the glow visual implementation uses CSS box-shadow or equivalent rendering primitive with color, blur-radius, and opacity parameters corresponding to each tier, ensuring the glow effect is purely additive (composites over standard item rendering) and does not affect item position, size, or tap/click target area.

Claim 20.4 (Glow Mechanic — Dependent)

The system of 20.1 wherein the cooperative-class Heart of Peace access guarantee is enforced at the data layer: inbox query delivers all OPEN questions to all members regardless of bounty tier; glow tier assignment is a presentation-layer transformation applied after data retrieval, ensuring that any presentation-layer failure (e.g., CSS rendering error) degrades gracefully to standard non-glowing inbox presentation without hiding any questions.

Claim 20.5 (Glow Mechanic — Dependent: Independent Method Claim)

A method for cooperative attention-surfacing measurement comprising: (a) presenting two cohorts of cooperative members with identical question sets, one cohort using the Glow Mechanic and one using standard flat-list presentation; (b) measuring mean time-to-first-response for questions grouped by Marks bounty tier in each cohort; (c) computing response-acceleration ratio per tier as (flat-list mean TTR) / (glow-mechanic mean TTR); and (d) recording the ratio per tier in the cooperative network analytics substrate as a glow-efficacy metric.

Empirical Receipt and Reduction to Practice

Glow Mechanic was ratified during BP083 Plow Cycle as the cooperative-class attention-surfacing primitive for the Package Store and Diagnosis cooperative inbox. Five-tier scale (0 / 1-9 / 10-99 / 100-999 / 1000+ Marks) was ratified by Founder. The access-preservation guarantee distinguishes this mechanism from conventional notification tier systems that gate visibility.

Claim Group 21: Three-Salt-Layer Architecture

Background and Problem Statement

Prior cooperative AI systems treat knowledge retrieval as a single-layer operation: query the local knowledge base or query the cloud API. No prior system provides a named, composable three-layer architecture that routes an

unanswered question through progressively broader knowledge sources—local verified substrate, cross-machine constellation, and human cooperative network—with each layer providing a distinct answer quality guarantee and escalating automatically when the prior layer cannot achieve concordance.

Detailed Description

Claim 21.1 (Three-Salt-Layer Architecture — Independent Claim)

A computer-implemented three-layer cooperative AI answer-finding architecture comprising: - (a) Layer 1 — Substrate Salt: the local verified-facts layer, implemented as the active Substrate directory (Claim Group 7) populated by the canonical plow pipeline (Claim Group 1); queries are resolved at this layer if a CONCORDANT or PARTIAL eplet with BMV above the configurable minimum threshold exists for the question domain and topic; Layer 1 resolution produces an answer with provenance class LOCAL-VERIFIED; - (b) Layer 2 — Federation Salt: the cross-machine constellation query layer, implemented as Tier 2 of the Federated Andon 3-Tier Escalation (Claim Group 17); queries escalate to this layer when Layer 1 fails to produce a CONCORDANT result within the Andon retry cap; Layer 2 resolution involves distributing the question to available Constellation peers (Claim Group 16 fleet) and applying concordance arbitration across peer-returned candidates; Layer 2 resolution produces an answer with provenance class CONSTELLATION-CONCORDANT; - (c) Layer 3 — Human Salt: the cooperative human-network query layer, implemented as The Diagnosis persistent broadcast system (Claim Group 18); queries escalate to this layer when Layer 2 fails to achieve concordance across all available Constellation peers; Layer 3 resolution produces an answer with provenance class HUMAN-CONCORDANT and writes the resolved answer back to Layer 1 (Substrate) for future local resolution.

Claim 21.2 (Three-Salt-Layer Architecture — Dependent)

The architecture of 21.1 wherein layer escalation is automatic and does not require user action at any escalation boundary; the Salt-Level persistence tier (Claim Group 19) configured by the user at question submission governs whether Layer 2 and Layer 3 escalation are permitted (A Pinch of Salt: Layer 1 only; Seasoning: Layers 1-2; Preserved in Salt: Layers 1-3).

Claim 21.3 (Three-Salt-Layer Architecture — Dependent)

The architecture of 21.1 wherein Layer 3 Human Salt write-back (HUMAN-CONCORDANT answer to Substrate) ensures that the three-layer architecture is self-improving: each human-resolved Diagnosis reduces future Layer 3 escalation frequency for similar questions, progressively strengthening the Layer 1 substrate.

Claim 21.4 (Three-Salt-Layer Architecture — Dependent)

The architecture of 21.1 wherein each layer's answer provenance class (LOCAL-VERIFIED / CONSTELLATION-CONCORDANT / HUMAN-CONCORDANT) is stored in the eblet metadata and surfaced in the AI model's response attribution, enabling the user to assess the epistemic source of each answer.

Claim 21.5 (Three-Salt-Layer Architecture — Dependent: Independent Method Claim)

A method for three-layer cooperative architecture efficacy measurement comprising: (a) recording per-question resolution layer (L1/L2/L3) and provenance class in the substrate receipt; (b) computing the L1-resolution rate as the fraction of questions resolved at Layer 1 across a canonical benchmark run; (c) computing the L2-escalation rate and L3-escalation rate correspondingly; and (d) tracking L1-resolution rate trend across successive benchmark runs, wherein an increasing L1-resolution rate demonstrates substrate self-improvement via Human Salt write-back.

Empirical Receipt and Reduction to Practice

Three-Salt-Layer Architecture was ratified during BP083 Plow Cycle as the canonical composite architecture combining the Local Substrate (BP083 Plow Pipeline receipt: 68/70 MMLU-Pro), the Constellation Federation (MIC 5-machine fleet), and The Diagnosis human network. The three-salt naming convention (Substrate Salt / Federation Salt / Human Salt) directly maps to the Salt-Level Persistence-Tier Selector (Claim Group 19) nomenclature, creating a unified metaphor across the user-facing selection interface and the system-internal architecture.

Claim Group 22: Hardware-Tier Model Selection

Background and Problem Statement

Consumer hardware available in the 2024-2026 era exhibits a wide range of RAM capacities—commonly 8 GB, 16 GB, 32 GB, and 48+ GB—that directly determines which local AI model weight classes can be loaded without triggering silent out-of-memory failure or operating system memory pressure events. No prior cooperative AI deployment system provides an automated RAM detection mechanism that maps detected RAM to a recommended model tier, presents this recommendation to the user at first launch, and enforces the recommendation as a default model selection to prevent silent OOM failure.

The empirical motivation is a confirmed Founder incident wherein 8 GB consumer hardware silently OOM-failed when loading gemma4:12b, requiring a manual workaround alias redirect to a lower-parameter model. The Hardware-Tier Model Selection system obsoletes the workaround alias by making the appropriate model selection automatic and transparent.

Detailed Description

Claim 22.1 (Hardware-Tier Model Selection — Independent Method Claim)

A computer-implemented method for cooperative-class AI hardware-tier model selection comprising: - (a) a RAM detection phase executed at application first launch and model selection UI open, querying the operating system for total installed physical RAM via platform-appropriate system API (Windows: GlobalMemoryStatusEx; macOS: sysctl hw.memsize; Linux: /proc/meminfo MemTotal); - (b) a tier classification function mapping detected RAM to one of four model tiers: — Lightweight (~6 GB VRAM/RAM footprint): recommended at detected RAM < 10 GB; default model: gemma2:2b or equivalent sub-6B-parameter quantized model; — Standard (~12 GB footprint): recommended at detected RAM >= 10 GB and < 20 GB; default model: gemma4:12b or equivalent 10-13B-parameter model; — Premium (~14 GB footprint): recommended at detected RAM >= 20 GB and < 40 GB; default model: gemma4:14b or equivalent 13-16B-parameter model; — Heavy (~48 GB footprint): recommended at detected RAM >= 40 GB; default model: gemma3:27b or equivalent 27B+ parameter model; - (c) a recommendation display that presents the detected RAM value, computed tier classification, recommended model identifier, and estimated RAM footprint to the user at first-launch model selection with a plain-English explanation of why this tier is recommended; - (d) a user-override mechanism allowing the user to select a higher or lower tier than recommended, with a displayed warning if the user selects a tier whose RAM footprint exceeds 80% of detected RAM; and - (e) an automatic model-pull directive issuing the appropriate Ollama pull command (or equivalent local model manager command) for the recommended or user-selected model tier, displaying a progress indicator per cooperative-class long-running progress discipline.

Claim 22.2 (Hardware-Tier Model Selection — Dependent)

The method of 22.1 wherein the RAM detection result is stored in the application configuration alongside the user's tier selection and model identifier, enabling the application to detect post-reinstall hardware changes by comparing stored vs. current RAM and re-presenting the tier recommendation if a material change (default: >= 4 GB delta) is detected.

Claim 22.3 (Hardware-Tier Model Selection — Dependent)

The method of 22.1 wherein the tier classification function composes with the Onboarding Lifecycle Auto-Advance mechanism (Claim Group 23) such that a first-launch hardware-tier recommendation is presented as part of the onboarding Stage A flow, and successful model pull completion advances the onboarding stage machine from A to B without user action.

Claim 22.4 (Hardware-Tier Model Selection — Dependent)

The method of 22.1 wherein the workaround-alias antipattern—manually redirecting model requests from a requested model identifier to a lower-parameter model alias—is replaced by the tier classification function: if a user-

requested model identifier's estimated RAM footprint exceeds 80% of detected RAM, the application presents a tier-downgrade recommendation before initiating the model pull, rather than silently redirecting at inference time.

Empirical Receipt and Reduction to Practice

Hardware-Tier Model Selection was motivated by Founder-direct empirical observation: 8 GB machine silently OOM-failed on gemma4:12b; manual workaround alias was required to redirect to a lower-tier model. The tier boundaries (Lightweight/Standard/Premium/Heavy) were derived from typical quantized model RAM footprints for the 2024-2026 consumer hardware era. Reduction to practice: tier classification function implemented as cooperative-class first-launch flow during BP083 Plow Cycle development.

Claim 22.5 (Hardware-Tier Model Selection — Dependent Claim)

(dependent on 22.1) The system of 22.1 wherein the cooperative workspace includes a shared annotation layer allowing multiple members to attach commentary to shared documents without modifying the source record.

Claim Group 23: Plow Resume from Checkpoint

Background and Problem Statement

The canonical plow pipeline (Claim Group 1) executed on a full 14-domain MMLU-Pro benchmark may require hours of wall-clock time on consumer hardware. Prior plow pipeline implementations restart from question 1 upon any interruption: application crash, system reboot, user-initiated termination, or power event. This produces total work loss proportional to pipeline progress at time of interruption, with no mechanism for partial result preservation or selective restart.

No prior cooperative AI substrate pipeline provides periodic checkpoint state persistence, interrupt detection at relaunch, or automatic resume from the last persisted checkpoint.

Detailed Description

Claim 23.1 (Plow Resume from Checkpoint — Independent Method Claim)

A computer-implemented method for interrupt-resilient cooperative AI substrate pipeline execution comprising: - (a) a periodic checkpoint writer that, at a configurable interval (default: after every N questions, where $N = 5$), writes the current pipeline state to a named checkpoint file in the TestData directory, wherein the checkpoint file contains: current question index within the domain question bank, current domain identifier, per-domain question results accumulated to date (answer, BMV score, concordance class, Andon count, and eblet count), Andon state for the current question (retry count, widened operator set), and a SHA-256 hash of the checkpoint content for integrity verification; - (b) a checkpoint presence detector executed at every pipeline run launch that scans the TestData directory for checkpoint files matching the current domain

and benchmark run identifier; - (c) a checkpoint resume prompt that, upon detecting a valid checkpoint file (hash verification passes), presents to the user: the interrupted domain, the question index at interruption, the elapsed questions count and remaining questions count, and a binary choice (Resume from checkpoint / Restart from question 1); - (d) a resume-from-checkpoint execution path that, if the user selects resume: reads the checkpoint's accumulated results into the in-memory pipeline state, sets the starting question index to $\text{checkpoint_question_index} + 1$, and begins pipeline execution at that index without replaying prior questions; and - (e) a checkpoint cleanup mechanism that deletes the checkpoint file upon successful completion of the domain pipeline run, preventing stale checkpoints from interfering with subsequent runs.

Claim 23.2 (Plow Resume from Checkpoint — Dependent)

The method of 23.1 wherein checkpoint file naming encodes the domain identifier, benchmark run timestamp, and question count to enable disambiguation of multiple interrupted runs across domains in the same TestData directory.

Claim 23.3 (Plow Resume from Checkpoint — Dependent)

The method of 23.1 wherein the checkpoint writer composes with the Per-Domain Isolation Architecture (Claim Group 5) such that each domain maintains an independent checkpoint file; a system interruption during multi-domain parallel MIC-orchestrated execution (Claim Group 16) produces per-domain checkpoint files enabling selective resume of interrupted domains without re-running already-completed domains.

Claim 23.4 (Plow Resume from Checkpoint — Dependent)

The method of 23.1 wherein the SHA-256 checkpoint integrity verification detects file corruption, partial writes (from power-loss mid-write), and unauthorized modification; a checkpoint failing hash verification is treated as absent and the user is prompted to restart from question 1 with a notification that the checkpoint was corrupted.

Claim 23.5 (Plow Resume from Checkpoint — Dependent)

The method of 23.1 wherein the Andon state preservation in the checkpoint enables correct Andon retry count restoration upon resume: if the interrupted question had consumed K of the maximum Andon retry count, the resumed pipeline applies the remaining ($\text{max_retry} - K$) retries, preventing an interrupted Andon sequence from receiving more total retries than a non-interrupted sequence.

Empirical Receipt and Reduction to Practice

Plow Resume from Checkpoint was designed during BP083 Plow Cycle to address total-work-loss risk on multi-hour canonical benchmark runs. The cooperative-class waste-avoidance discipline (Substitution canon: prior work

should be made more valuable, not discarded) motivates checkpoint persistence as a structural architecture primitive rather than an optional recovery feature.

Claim Group 24: Onboarding Lifecycle Auto-Advance

Background and Problem Statement

Cooperative AI application onboarding involves multiple sequential stages that require distinct user actions (model pull, substrate seeding, first inference test) and may be interrupted by system events (restart, network loss, model download failure) at any stage. Prior cooperative AI onboarding implementations provide no mechanism for detecting that a user is persistently stuck at a given stage, surfacing a recovery diagnostic, or automatically advancing a stage that completed without user awareness.

The empirical motivation is a Founder-direct report: the Founder's son's machine was stuck at Stage B of the cooperative onboarding flow for multiple days. Reinstallation did not cure the stuck state. The root cause was profile-persistence across reinstall: the onboarding state was stored in a location (e.g., user profile directory or registry key) that survived application reinstall, causing the reinstalled application to re-read a corrupt or incomplete Stage B state and re-present Stage B without explaining why or offering recovery.

Detailed Description

Claim 24.1 (Onboarding Auto-Advance — Independent System Claim)

A computer-implemented system for cooperative-class onboarding lifecycle management comprising: - (a) a three-stage onboarding lifecycle model: Stage A (hardware detection and model tier selection, culminating in successful model pull completion); Stage B (Starter Chocolate substrate installation and first substrate query verification); Stage C (first successful user question with substrate-enriched response, confirming end-to-end pipeline operation); - (b) a stage state persistence mechanism storing the current stage identifier, stage entry timestamp, and stage completion status in a clearly named application data file separate from user profile or system registry, such that reinstall of the application binary replaces the state file with a fresh Stage A state without reading prior stage state from surviving locations; - (c) a stuck-stage detector that at each application launch computes the elapsed time since stage entry for the current incomplete stage and compares it to a per-stage configurable threshold (default: Stage A 30 minutes; Stage B 60 minutes; Stage C 120 minutes); - (d) a reset-diagnostic surfacing mechanism that upon detecting a stuck stage: presents an inline (non-modal) notification at the top of the onboarding view stating the current stage, elapsed time, and a plain-English description of the most likely cause of the stuck state per stage; and presents a Reset and Retry button that clears the current stage state and re-initiates stage entry actions from the beginning of the current stage without reverting prior stages; - (e) an auto-advance mechanism that polls for stage completion conditions at a configurable interval (default: 30 seconds)

during active application runtime: Stage A completion condition = successful Ollama model list response containing the recommended model identifier; Stage B completion condition = successful substrate directory creation and eblet count > 0; Stage C completion condition = successful inference response with substrate context injected; upon detecting any completion condition, the lifecycle advances to the next stage and the onboarding view updates without user action; and - (f) a profile-persistence-across-reinstall detection mechanism that at Stage A entry checks for surviving onboarding state from a prior installation in known persistence locations (user home directory, registry, AppData) and, if found, presents a notification: “We found a previous installation state. Would you like to start fresh or resume?” with Resume and Start Fresh options.

Claim 24.2 (Onboarding Auto-Advance — Dependent)

The system of 24.1 wherein Stage B’s Starter Chocolate installation (Claim Group 8) is the primary completion action, and the auto-advance polling verifies both (i) the idempotency marker file presence and (ii) an eblet count query returning > 0 results, confirming successful substrate seeding before advancing to Stage C.

Claim 24.3 (Onboarding Auto-Advance — Dependent)

The system of 24.1 wherein the reset-diagnostic surfacing mechanism composes with the Hardware-Tier Model Selection (Claim Group 22) at Stage A: a Stage A stuck state longer than the threshold triggers a re-presentation of the hardware tier recommendation, checking whether the model pull may have silently OOM-failed, and offering to switch to the Lightweight tier model as a recovery option.

Claim 24.4 (Onboarding Auto-Advance — Dependent)

The system of 24.1 wherein all onboarding lifecycle UI state transitions produce visible cooperative-class feedback per every-click-visible-feedback discipline: stage advance produces a brief celebratory confirmation notification; stuck-stage detection produces the inline warning notification; reset produces a progress indicator for re-initiating stage actions.

Claim 24.5 (Onboarding Auto-Advance — Dependent)

The system of 24.1 wherein Stage C completion produces a first-inference receipt: the question text, the substrate eblets consulted, the answer produced, and the inference latency are recorded to a first-inference-receipt file in the Vault directory, timestamped and SHA-256 hashed, providing the user with a permanent record of cooperative AI substrate operation.

Empirical Receipt and Reduction to Practice

Onboarding Lifecycle Auto-Advance was designed in direct response to Founder-direct empirical report: son’s machine stuck at Stage B for multiple days; reinstall did not cure. Root cause: profile-persistence of onboarding state across reinstall.

Reduction to practice: three-stage lifecycle model with per-stage thresholds, reset-diagnostic surfacing, auto-advance polling, and profile-persistence-across-reinstall detection implemented during BP083 Plow Cycle architecture session.

Claim Group 25: Constellation Switchboard NotCents Shape Mapping

Background and Problem Statement

Prior cooperative network status displays use single-dimension indicators (online/offline, green/red) that cannot communicate the combination of connectivity state, activity state, and role assignment in a compact at-a-glance visual primitive. A cooperative peer fleet display requires communicating at minimum four distinct states (connected-idle, connected-active, offline, and unknown/connecting) plus a Conductor-role designation for the MIC-assigned orchestrating machine, all within the compact confines of a constellation network topology view.

No prior system provides a named three-shape four-state visual primitive with a composable Conductor-role modifier specifically designed for cooperative AI peer fleet status communication.

Detailed Description

Claim 25.1 (Constellation Switchboard — Independent System Claim)

A computer-implemented visual system for cooperative peer fleet status display comprising a three-shape four-state visual primitive applied per-peer in a Constellation Switchboard UI view: - (a) Square shape (□) — represents the connected-idle state: the peer is reachable via health-check, is a registered Constellation member, and has no active domain pipeline workload assigned; - (b) Triangle shape (△) — represents the connected-active state: the peer is reachable and currently executing an assigned domain pipeline workload (MIC partition assignment active, or local plow pipeline running); - (c) Circle shape (○) — represents the offline state: the peer's last health-check response exceeded the configurable timeout threshold (default: 3 missed polls) or the peer has not yet been reached since application launch; wherein each shape is further modified by a fill state: - (d) Filled shape (solid interior, e.g., ■ ▲ ●) — represents the lit state: the peer's most recent health-check succeeded within the current polling cycle; - (e) Outline shape (border only, e.g., □ △ ○) — represents the unlit state: the peer has not responded to the most recent health-check cycle (but has not exceeded the offline threshold).

Claim 25.2 (Constellation Switchboard — Dependent)

The visual system of 25.1 further comprising a Conductor-role modifier applied to the shape of the peer currently designated as the MIC Conductor-class orchestrator (Claim Group 16): - (a) a crown overlay rendered above the shape's bounding box; or - (b) a filled-star (★) modifier replacing the fill indicator of the shape; wherein only one peer may display the Conductor-role modifier at any time, reflecting the single active MIC Conductor designation.

Claim 25.3 (Constellation Switchboard — Dependent)

The visual system of 25.1 wherein the full four-state two-shape-modifier matrix produces eight distinct visual states per peer: connected-idle-lit (■□ filled square), connected-idle-unlit (□ outline square), connected-active-lit (▲ filled triangle), connected-active-unlit (△ outline triangle), offline-lit (● filled circle), offline-unlit (○ outline circle), connecting-lit (■ filled square with animated border), connecting-unlit (□ outline square with animated border); with the Conductor modifier compositing over any of the eight states.

Claim 25.4 (Constellation Switchboard — Dependent)

The visual system of 25.1 wherein shape selection (square/triangle/circle) is determined by the most recent activity state reported in the peer's health-check response payload, and fill state (filled/outline) is determined by health-check recency within the current polling cycle, ensuring that connectivity events (peer goes offline) are reflected in shape change and polling recency events (poll missed but peer not yet timed out) are reflected in fill change, providing two independent visual channels for two independent status dimensions.

Claim 25.5 (Constellation Switchboard — Dependent)

The visual system of 25.1 wherein peer shape and fill state are recomputed at each health-check polling cycle and applied to the Constellation Switchboard view via a reactive UI binding, ensuring the display reflects current fleet status without page reload or manual refresh.

Claim 25.6 (Constellation Switchboard — Dependent: Independent Method Claim)

A method for cooperative fleet status communication comprising: rendering the three-shape four-state visual primitive for each configured Constellation peer in a grid or topology view; compositing the Conductor-role modifier on the designated MIC peer; updating fill state at each polling cycle; updating shape state upon activity-state change events received from peer health-check payloads; and providing a legend view accessible via tap/click on any shape explaining the full eight-state matrix in plain language.

Empirical Receipt and Reduction to Practice

Constellation Switchboard NotCents Shape Mapping was ratified during BP083 Plow Cycle by Founder direct: Square=idle-connected, Triangle=active, Circle=offline, filled/outline=lit/unlit, Conductor-crown overlay. The four-state two-modifier design was validated against a 5-machine Constellation fleet (M0-M4) providing sufficient state expressiveness for cooperative peer fleet operational awareness.

Claim Group 26: Substitution Rail Cooperative Marketplace

Background and Problem Statement

Prior digital knowledge marketplaces (a) require fiat currency for all transactions, gating access to users with payment methods; (b) provide no cooperative alternative currency exchange mechanism; (c) do not integrate provenance-tracking medallion stamps on purchased knowledge units; or (d) impose a platform take-rate that extracts value from creators without cooperative distribution to all participants.

The Substitution Rail Cooperative Marketplace extends the Substitution Rail Triple-Mode Exchange (Claim Group 12) with a full cooperative substrate-eblet pack marketplace architecture implementing three payment rails operating simultaneously on the same catalog, cooperative-class creator distribution, Cross-Pollination Reciprocity at-risk acquisition for high-tier packs, and Medallion IP provenance stamps on every installed pack unit.

Detailed Description

Claim 26.1 (Substitution Rail Marketplace — Independent System Claim)

A computer-implemented cooperative substrate-eblet pack marketplace comprising: - (a) a catalog layer indexing available knowledge packs by domain, tier (Bread full-corpus / Chocolate curated / Kitchen custom), creator cooperative node identifier, reputation score, and Marks price; - (b) a three-rail payment dispatch that at transaction time routes payment through exactly one of three Substitution Rails selected by the purchasing member: — Fiat Pipeline Rail: conventional currency transaction at Cost+20%; the Liana Banyan cooperative pipeline collects the Fiat transaction and distributes to the creator cooperative node at the cooperative's creator-distribution ratio (83.3% to creators per cooperative bylaw), with the pipeline earning Cost+20% minus the creator distribution as operating margin; at no point does the cooperative hold Fiat funds beyond the transaction settlement window; — Marks Rail: cooperative credit transaction wherein the purchasing member's Marks balance is decremented by the listed Marks price and the creator cooperative node's Marks balance is incremented by the same amount; no fiat exchange occurs; cooperative currency circulates exclusively within the cooperative; — Barter Rail: direct peer-to-peer exchange recorded as a barter contract wherein the purchasing member pledges goods, services, or knowledge units of equal or greater cooperative-assessed value; the barter contract is written to the cooperative ledger with both parties' signatures and a cooperative-assessed valuation; - (c) a Cross-Pollination Reciprocity (XP/RP) mechanism applying to packs priced at or above a configurable Marks threshold (default: 5,000 Marks): the purchasing member must place a Marks stake of equal value at risk in the cooperative ledger at transaction time; the stake is held for a configurable escrow period (default: 30 days); if the purchasing member publishes a cooperative knowledge pack within the escrow period with reputation-weighted reception above the configurable

minimum, the stake is returned; otherwise the stake is distributed to the cooperative general fund; - (d) a Medallion IP provenance stamp mechanism that at pack installation time: generates a unique Medallion identifier for each installed pack unit; writes the Medallion identifier, originating cooperative node identifier, installation timestamp, and purchasing member identifier to the cooperative IP ledger; and tags each eblet installed from the pack with the Medallion identifier in the eblet metadata; and - (e) a proximity and reputation tier display in the catalog view presenting each pack creator's cooperative reputation score, distance (in Constellation topology hops) from the purchasing member's node, and verified install count.

Claim 26.2 (Substitution Rail Marketplace — Dependent)

The marketplace of 26.1 wherein the Fiat Pipeline Rail's structural prohibition on fiat holding is enforced at the transaction validation layer: if the cooperative ledger detects a Fiat balance accumulation above a configurable maximum hold duration (default: 24 hours), the ledger flags the transaction for immediate settlement processing, ensuring the cooperative pipeline operates as a pass-through, not a financial institution holding member funds.

Claim 26.3 (Substitution Rail Marketplace — Dependent)

The marketplace of 26.1 wherein cooperative currency (Marks, Joules, Credits) received through any of the three Substitution Rails is prohibited from fiat conversion at the transaction validation layer: any transaction that would result in cooperative currency being exchanged for fiat currency is rejected with an error referencing the Substitution canon and cooperative bylaw.

Claim 26.4 (Substitution Rail Marketplace — Dependent)

The marketplace of 26.1 wherein the Medallion provenance stamp composes with the Catacombs Cold Archive (Claim Group 20) and Excalibur Active Currents pheromone decay (Claim Group 14): Medallion-stamped eblets receive a provenance-premium in EAC decay scoring, decaying more slowly than non-Medallion eblets of equal age and access frequency, reflecting the higher cooperative value represented by a purchased pack unit.

Claim 26.5 (Substitution Rail Marketplace — Dependent)

The marketplace of 26.1 wherein the Package Store Confectionary Architecture (Claim Group 11) provides the stall taxonomy (Bakery/Confectionary/Kitchen) and the Substitution Rail Cooperative Marketplace provides the payment and provenance infrastructure, composing into a complete cooperative knowledge distribution and exchange system.

Empirical Receipt and Reduction to Practice

Substitution Rail Cooperative Marketplace was ratified during BP083 Plow Cycle as the payment architecture for the Package Store Confectionary. Three payment rails (Fiat-Pipeline / Marks / Barter), 83.3% creator distribution ratio, Cross-Pollination Reciprocity at 5,000 Marks threshold, and Medallion IP

provenance stamp were all ratified by Founder. The structural fiat-conversion prohibition (cooperative currencies never convert to fiat) is enforced at the transaction validation layer per Substitution canon.

Claim Group 27: Tab UX Cooperative Interface Architecture and Test-Net by Design Ledger Architecture

Background and Problem Statement

Cooperative AI substrate applications present two distinct user populations: Regular members who consume cooperative services, and Advanced members who configure the substrate infrastructure. Prior tabbed UI architectures provide no mechanism for a single-row Regular/Advanced toggle with per-tab pin and hide controls that persist across sessions without triggering a context reload. Prior test-ledger implementations relying on deployment flags cannot demonstrate that the structural fiat-conversion prohibition is architecturally enforced rather than policy-enforced.

Detailed Description

Claim 27.1 (Tab UX + Test-Net by Design — Independent System Claim)

A computer-implemented cooperative-class tabbed interface system comprising: - (a) a single-row tab bar rendering all cooperative application tabs in a persistent horizontal row without vertical expansion; - (b) a Regular/Advanced toggle control switching between a Regular tab subset and an Advanced tab subset within the same horizontal row; - (c) a per-tab pin mechanism enabling any member to designate any tab as persistently pinned, wherein pinned tabs remain visible regardless of toggle state, and pin state persists across sessions in the member's cooperative account preferences; - (d) a per-tab hide mechanism enabling any member to mark any tab as hidden, removing it from the visible tab bar without deleting the underlying tab state, with a hidden-tab count indicator providing a restore pathway; - (e) a toggle-state persistence mechanism that records the member's most recent toggle selection and applies it at session start; and - (f) a tab context preservation guarantee wherein switching between Regular and Advanced states does not reload any currently displayed tab, preserving in-flight substrate queries, open forms, and accumulated session state.

Claim 27.2 (Tab UX + Test-Net by Design — Dependent)

The system of 27.1 wherein the Regular tab subset presents member-facing cooperative participation tabs for substrate query, marketplace, account management, and diagnosis inbox, and the Advanced tab subset extends the Regular subset with infrastructure tabs for pipeline monitoring, Constellation fleet view, Andon cord diagnostics, and substrate administration, such that a Regular member operates within a simplified four-to-six-tab interface while an Advanced member accesses the full twelve-to-eighteen-tab interface in the same single-row bar.

Claim 27.3 (Tab UX + Test-Net by Design — Dependent)

The system of 27.1 wherein the per-tab pin and hide state is recorded in a cooperative member interface audit log enabling the cooperative to identify interface improvement opportunities: persistent hide events on any tab by greater than or equal to 5% of active members trigger a relevance-mismatch recommendation; persistent pin events on any non-default tab by greater than or equal to 10% of active members trigger a promotion recommendation.

Claim 27.4 (Tab UX + Test-Net by Design — Dependent: Independent System Claim)

A computer-implemented cooperative-class ledger architecture for structurally isolated test-environment verification of cooperative currency mechanics comprising: - (a) a Test-Net ledger partition whose cooperative currency units carry a structural prefix distinguishing them from production units at the protocol layer, wherein the production ledger rejects any Test-Net-denominated currency in a production transaction regardless of deployment configuration or operator override; - (b) a fiat-conversion-prohibition structural enforcement layer in the Test-Net partition enforcing the same structural prohibition as in the production ledger, such that fiat-conversion prohibition testing validates the structural mechanism rather than merely the policy declaration; - (c) a Test-Net reset mechanism enabling authorized operators to reset Test-Net balances and records to a configurable initial state without affecting any production record; and - (d) a cryptographic namespace segregation mechanism assigning Test-Net record identifiers a structurally distinct namespace from production identifiers, preventing any Test-Net record from being referenced in a production context.

Claim 27.5 (Tab UX + Test-Net by Design — Dependent)

The architecture of 27.4 wherein the cryptographic namespace segregation composes with the Substitution Rail Triple-Mode Exchange (Claim Group 12) such that Test-Net cooperative currency may traverse the Fiat Pipeline Rail, Marks Rail, and Barter Rail test pathways, and the Fiat Pipeline Rail structural prohibition on fiat holding is enforced identically in both partitions, ensuring structural prohibition testing generates empirical receipts demonstrating the prohibition as an architectural constraint.

Claim 27.6 (Tab UX + Test-Net by Design — Dependent)

The architecture of 27.4 wherein the Test-Net partition composes with the Substitution Rail Cooperative Marketplace (Claim Group 26) by providing a structurally isolated test marketplace in which the Cross-Pollination Reciprocity stake-and-escrow mechanism, the Medallion IP provenance stamp generation, and the 83.3% creator distribution calculation can each be end-to-end verified before production deployment.

Empirical Receipt

Tab UX Option B (Item 22) and Test-Net by Design (Item 23) both ratified BP083. Pin/hide resolves attention-overload. Test-Net extends fiat-conversion-prohibition coverage to architectural verification.

Claim Group 28: Truth Integrity Chain (TIC) — Structured Epistemic Knowledge Representation

Anecdote — Founder Voice (verbatim from canon: founders_anecdote_africa_mother_dog_bp084)

When I was 3 years old, I was attacked by a dog in Africa. We were missionaries' children, and Africa was home for a season. I don't remember everything about that day, but I remember the dog — a mother protecting her puppies. She didn't know me.

Background and Problem Statement

The lesson of the Africa anecdote is not that logic is unreliable. The lesson is that logical validity is not factual soundness. A perfect Bayesian reasoner operating on incomplete data will reach wrong conclusions with complete confidence. The mother dog had no ELIMINATED category. She had no way to represent “that creature is NOT a threat.” She acted on the full set of what she knew, which was not the full set of what was true.

The language models that power most artificial intelligence products today have the same structural gap. They reason well. Their predictions are calibrated. Their internal logic is sophisticated. But they have no ELIMINATED category. When queried, these systems produce a single confident-sounding sentence. That sentence has no mechanism to surface: “This claim has been tested and found false.” It has no native representation of: “Here are twelve competing theories that have not been ruled out, each with different consequence implications.” It collapses uncertainty into a single output — exactly as the mother dog collapsed “small creature approaching puppies” into “attack.”

Prior approaches to improving artificial intelligence accuracy do not address this structural gap. Retrieval-augmented generation (RAG) adds more known facts to model context but carries no record of what has been ruled out. Fine-tuning modifies model weights but does not introduce an ELIMINATED field into the knowledge representation. Larger parameter counts improve prediction accuracy but do not change the epistemic structure: all three approaches continue to produce a single confident sentence without surfacing the space of eliminated hypotheses or the survival scores of competing open theories. No prior cooperative or non-cooperative artificial intelligence substrate provides structured, first-class, computationally-linked representation of: (a) verified known facts with evidence trails; (b) competing open theories with survival

scores; and (c) definitively falsified hypotheses with contradiction trails — all three classes maintained in a dependency graph that propagates updates when foundational knowledge changes.

The Truth Integrity Chain (TIC) addresses this gap by extending the cooperative-class eblet knowledge record with a five-field epistemic structure that makes ELIMINATED a first-class data type equal in tractability to KNOWN.

Detailed Description

1. The Five-Field Eblet Truth-Integrity Record

Each eblet in the cooperative-class knowledge substrate is extended with five structured fields constituting the TIC record.

Field 1 — KNOWN

Verified propositions with supporting evidence trails. Each KNOWN entry contains: (a) the proposition text; (b) one or more evidence-trail records, each comprising source identifier, retrieval timestamp, and supporting excerpt; (c) a confidence score derived from the Banyan Metric Value concordance arbitration (Claim Group 4); and (d) a chronos-stamp recording when the proposition achieved KNOWN status.

Example (gravity domain):

```
KNOWN:
- proposition: "Gravity propagates at the speed of light (c)"
  evidence_trail:
    - source: "LIGO-2017-GW170817"
      timestamp: "2017-08-17"
      excerpt: "GW and EM counterpart arrival delta < 1.7s over 130 Mpc"
    confidence: 0.97
    chronos: "2026-06-16T00:00:00Z"
- proposition: "Gravity attracts mass to mass in proportion to M1*M2/r^2"
  evidence_trail:
    - source: "Newton-Principia-1687"
    - source: "Einstein-GR-empirical-1915"
    confidence: 0.999
    chronos: "2026-06-16T00:00:00Z"
```

Field 2 — THEORIES_OPEN

Competing hypotheses that have not been eliminated. Each THEORIES_OPEN entry contains: (a) the theory text; (b) a survival score (initialized at 1.0, decremented by failed consequence probes, incremented by confirmed consequence probes); (c) a consequence-trace log recording all child queries spawned by Loop 10 and their outcomes; and (d) a chronos-stamp.

Field 3 — ELIMINATED

Hypotheses definitively falsified by evidence. Each ELIMINATED entry contains: (a) the hypothesis text; (b) the contradicting evidence record; (c) the eliminator identifier (Code Breaker member or pipeline loop identifier); and (d) a chronos-stamp.

Example (gravity domain):

```
ELIMINATED:
- hypothesis: "Gravity acts only between objects in direct physical
  proximity"
  contradicting_evidence:
    source: "Cavendish-1798-torsion-balance"
    excerpt: "Gravitational attraction between separated lead
      spheres"
  eliminated_by: "Loop-11-ELIMINATION_VERIFICATION"
  chronos: "2026-06-16T00:00:00Z"
- hypothesis: "Gravity propagates instantaneously"
  contradicting_evidence:
    source: "LIGO-2017-GW170817"
  eliminated_by: "Loop-11-ELIMINATION_VERIFICATION"
  chronos: "2026-06-16T00:00:00Z"
```

Field 4 — DEPENDENCIES_UPSTREAM

Directed graph edges to antecedent knowledge records on which this eblet's KNOWN propositions depend. Each entry contains: upstream eblet identifier, upstream KNOWN field key relied upon, and dependency-type label (foundational / supportive / compositional).

Field 5 — APPLICATIONS_DOWNSTREAM

Directed graph edges to derived knowledge records that depend on this eblet's KNOWN propositions. Each entry contains: downstream eblet identifier, downstream field that depends on this KNOWN, and a needs-reeval flag (Boolean) set to true by Loop 12 when upstream knowledge is updated.

2. The Hasty-Generalization Structural Guard

Universal-quantified claims — propositions containing “all,” “every,” “always,” “never,” or equivalent universal quantifiers — are subject to a structural guard that prevents premature promotion to KNOWN status. The guard comprises two requirements that must both be satisfied before a universally-quantified proposition may enter the KNOWN field:

(a) Minimum Sample Size Threshold. The evidence trail must contain a configurable minimum number of independent supporting sources (default: $N_{\min} = 5$ for universal claims; $N_{\min} = 1$ for existential claims). Any universally-quantified proposition with fewer than $N_{\min}$ sources is held in THEORIES_OPEN regardless of individual source confidence scores.

(b) Counter-Evidence Search Radius. The cooperative pipeline executes a counter-evidence search — a targeted query for evidence contradicting the universally-quantified claim — across a configurable radius of specialist sources (default: all available specialist adapters per Claim Group 2). If the counter-evidence search yields a result with contradiction confidence above a configurable threshold (default: 0.70), the proposition is held in THEORIES_OPEN with the counter-evidence cited, and a Code Breaker review is surfaced (Claim Group 29).

The hasty-generalization guard is activated automatically by the Scribe stage (Claim Group 1, element (h)) whenever Three Fates concordance arbitration produces a CONCORDANT or PARTIAL result containing a universal quantifier in the proposition text.

3. The Chronos-Linked Dependency Graph

DEPENDENCIES_UPSTREAM and APPLICATIONS_DOWNSTREAM collectively constitute a directed acyclic graph (DAG) spanning the cooperative eblet corpus. Every edge carries a chronos-stamp recording when the dependency relationship was established. The DAG enables two operations:

Dependency Traversal. When a user queries the basis for a KNOWN proposition, the dependency graph is traversed upstream, producing a human-readable provenance trail from the queried proposition back to foundational sources. **Propagation on Update.** When a KNOWN entry is updated, the dependency graph is traversed downstream through APPLICATIONS_DOWNSTREAM edges, flagging each dependent eblet with needs_reeval: true.

4. The 12-Loop Plow Extension

— Loops 10, 11, and 12

The canonical nine-stage plow pipeline (Claim Group 1) is extended with three loops that operate after the Scribe stage (element (h)) on each successfully minted eblet. **Loop 10 — CONSEQUENCE_TRACE**

For each entry T in the eblet's THEORIES_OPEN field, Loop 10 identifies a configurable number of immediate logical consequences of T (default: k=3 consequences). For each consequence C, Loop 10 executes the canonical plow pipeline as a child query. If the child eblet's KNOWN field contains a proposition consistent with C, T's survival_score is incremented (default: +0.05).

5. Worked Example

— Gravity Claims

Eblet ID: gravity-classical-attraction-K001

```

KNOWN:
  "Gravity attracts mass to mass in proportion to  $M1 \cdot M2 / r^2$ "
  (confidence: 0.999)
  "Gravitational waves propagate at the speed of light" (confidence:
  0.97)

THEORIES_OPEN:
  "Gravity is mediated by gravitons (spin-2 massless bosons)"
  (survival_score: 0.82)
  "Gravity is an emergent thermodynamic phenomenon – Verlinde 2010"
  (survival_score: 0.64)
  "Modified Newtonian Dynamics (MOND) as dark matter alternative"
  (survival_score: 0.51)

ELIMINATED:
  "Gravity acts only between objects in direct physical proximity"
  (Loop-11)
  "Gravity is electromagnetic in origin" (Loop-11; Faraday 1850)
  "Gravity propagates instantaneously" (Loop-11; LIGO 2017)

DEPENDENCIES_UPSTREAM:
  newtonian-mechanics-K000 / mass-attraction / foundational

APPLICATIONS_DOWNSTREAM:
  orbital-mechanics-K002 / gravitational-force-formula
  black-hole-ringdown-K017 / GR-predicts-ringdowns
  gravitational-lensing-K031 / light-deflection-by-mass

```

Loop 10 on MOND: consequence probe “Does MOND predict galaxy cluster lensing correctly?” → substrate surfaces Bullet Cluster (2006) data contradicting MOND prediction → survival_score 0.64 decremented to 0.54. Probe “Does MOND predict dwarf galaxy rotation curves?” → confirmed → survival_score 0.54 incremented to 0.59. Theory remains OPEN (above 0.20 threshold) but weakened. Loop 11 on “Gravity propagates instantaneously”: substrate search finds KNOWN K-001 proposition “Gravitational waves propagate at c ” with confidence $0.97 \geq 0.70$ threshold → theory migrated to ELIMINATED; Code Breaker queue updated.

Figures

Diagram A — Five-Field TIC Record Block Diagram (Figure 20)

TIC EBLET TRUTH-INTEGRITY RECORD		
KNOWN	THEORIES_OPEN	ELIMINATED
(verified)	(open + survival	(falsified +
- proposition	scores)	contradiction trail)
- evidence_trail	- theory	- hypothesis
- confidence	- survival_score	- contradicting_evidence
- chronos	- consequence_trace	- eliminated_by
	- chronos	- chronos
DEPENDENCY GRAPH FIELDS		
DEPENDENCIES_UPSTREAM	APPLICATIONS_DOWNSTREAM	
(antecedent edges)	(derived edges)	
- upstream_id	- downstream_id	
- field	- field	
- type (foundational/ supportive/compositional)	- needs_reeval (bool)	
	- needs_reeval_chronos	

Diagram B — Consequence Chain (Figure 21)

```
Theory: MOND (survival: 0.64)
|
+-- Loop 10 consequence probe 1: "Does MOND predict cluster
lensing?"
|   Result: CONTRADICTION (Bullet Cluster 2006)
|   survival_score -= 0.10 --> 0.54
|
+-- Loop 10 consequence probe 2: "Does MOND predict dwarf galaxy
rotation?"
|   Result: CONFIRMED
|   survival_score += 0.05 --> 0.59
|
+-- Loop 10 consequence probe 3: "Does MOND predict CMB acoustic
peaks?"
|   Result: CONTRADICTION (CMB power spectrum inconsistent)
|   survival_score -= 0.10 --> 0.49
|   [Still above 0.20 threshold -- stays THEORIES_OPEN,
weakened]
```

Diagram C — Dependency Propagation (Figure 22)

```

KNOWN K-001: "Gravity propagates at c" [UPDATED]
|
+-- APPLICATIONS_DOWNSTREAM (Loop 12, depth=1):
|   orbital-mechanics-K002      [needs_reeval: true]
|   black-hole-ringdown-K017   [needs_reeval: true]
|   gravitational-lensing-K031 [needs_reeval: true]
|   gw-speed-K089              [needs_reeval: true]
|   binary-pulsar-timing-K112  [needs_reeval: true]
|
+-- depth=2 (from gravitational-lensing-K031):
|   cosmological-distance-K201  [needs_reeval: true]
|
+-- depth=3 (from cosmological-distance-K201):
    Hubble-constant-K315        [needs_reeval: true]
    [All flagged --> surfaced in cooperative review queue]

```

Diagram D — Hasty-Generalization Guard (Figure 23)

```

Three Fates produces CONCORDANT result
|
v
Does proposition contain universal quantifier? (all/every/always/never)
|
+-- NO --> Standard KNOWN promotion
|
+-- YES -->
    |
    v
    Evidence trail count >= N_min (default: 5)?
    |
    +-- NO --> Hold in THEORIES_OPEN (insufficient sample)
    |
    +-- YES -->
        |
        v
        Counter-evidence search across all specialist adapters
        |
        v
        Contradiction confidence >= 0.70?
        |
        +-- YES --> Hold in THEORIES_OPEN + Code Breaker
review
        |
        +-- NO --> Promote to KNOWN (universal-claim
flagged)

```

Liana Banyan Corporation Inventor: J. Jones * Provisional Patent Application BP084 * Sonnet 4.6 * June 16, 2026 Innovation Area 35 * Claim Group 28 * status: founder-ratify-pending*

Empirical Receipt

TIC ratified BP084. Anchors: Africa mother-dog anecdote (logical validity vs. factual soundness); Bob-is-a-dog formalization; Gravity Claims TIC (KNOWN: gravity at c; THEORIES_OPEN: graviton, MOND; ELIMINATED:

proximity-only, electromagnetic). Loops 10/11/12 extend CG1 canonical plow to make ELIMINATED a first-class data type.

Claims — Claim Group 28

28.1 (independent) — Truth Integrity Chain Five-Field Record

A system for structured epistemic knowledge representation comprising: a five-field truth-integrity record storing for each knowledge claim a KNOWN field, an OPEN field, an ELIMINATED field, a CONSEQUENCE field, and a SURVIVAL_SCORE field; a dependency graph engine linking each claim to its upstream premises; and a decay function that reduces the SURVIVAL_SCORE of any claim whose upstream premise is challenged or eliminated.

Claim 28.2 (Truth Integrity Chain — Dependent Claim)

(dependent on 28.1) The system of 28.1 wherein the Hasty-Generalization Guard rejects claim entries that lack a population-scope qualifier or assert universal quantification without an enumerated evidence source.

Claim 28.3 (Truth Integrity Chain — Dependent Claim)

(dependent on 28.1) The system of 28.1 wherein the Chronos-Linked Dependency Graph propagates SURVIVAL_SCORE updates from challenged premises to all downstream dependent claims within a configurable cascade depth.

Claim 28.4 (Truth Integrity Chain — Dependent Claim)

(dependent on 28.1) The system of 28.1 wherein the 12-Loop Plow Extension processes Loops 10 (Consequence Probe), 11 (Contradiction Detection), and 12 (Survival Score Recompute) as asynchronous substrate tasks triggered by any ELIMINATED-state event.

Claim 28.5 (Truth Integrity Chain — Dependent Claim)

(dependent on 28.1) The system of 28.1 wherein each five-field record is immutable after creation except via authorized Code Breaker Guild challenge events, and wherein all state transitions are logged with submitter identity, timestamp, and challenge resolution outcome.

Claim Group 29: Code Breakers Guild and Gold Refined by Fire

Anecdote — Founder Voice

What survives the smashing is Immutable. Code Breakers smash. Marks pay them. Gold Refined by Fire.

Here is the cooperative principle behind all of this: we pay people to break things. The Code Breakers Guild is an open cooperative role — anyone can join. Code Breakers take claims that have been submitted to the substrate and attempt to demonstrate that they are false, incomplete, or overstated.

Background and Problem Statement

The scientific replication crisis — documented extensively in the literature following Ioannidis (2005, “Why Most Published Research Findings Are False”) — demonstrates that traditional peer review is structurally inadequate as a verification mechanism. Unpaid reviewers have no financial stake in thorough adversarial review. Anonymous review removes attribution accountability. Confirmation bias in the reviewer pool — wherein researchers with sympathetic priors are selected for review — systematically favors confirmation over falsification. The result is a published literature in which the majority of findings in certain fields do not replicate under independent testing.

Bug bounty programs in software security represent the closest prior art with the correct incentive structure: adversarial parties (security researchers) are paid specifically to find failures in claims made by a system (“this software is secure”). Bug bounty programs have consistently outperformed unpaid internal security review in finding critical vulnerabilities, because financial incentives align the reviewer’s interests with thorough adversarial searching rather than cursory confirmation.

No prior cooperative knowledge system provides: (a) a tiered claim-advancement mechanism where claims progress from UNTESTED through TESTED and FORGED to an IMMUTABLE designation only upon surviving a configurable number of adversarial challenges; (b) a named cooperative member tier progression that rewards members specifically for successful falsification work, equal in economic dignity to affirmation work; (c) a Refiner of Gold honor that inverts the typical recognition structure by awarding the highest cooperative distinction to members whose honest hardest attacks failed — because those failed attacks are the direct cause of a claim achieving Immutable status; (d) a tagged cooperative currency denomination for falsification work, enabling the cooperative to track and balance the ratio of affirmation work to falsification work across its member base; or (e) a reversal mechanism for Immutable claims that requires new empirical evidence plus cooperative majority vote, establishing a structurally high bar that is occasionally crossed.

Detailed Description

1. The Four-Tier Knowledge Claim Progression

Knowledge claims within the cooperative substrate advance through four tiers based on accumulated verification evidence. The tiers are mutually exclusive: a claim occupies exactly one tier at any moment.

Tier 1 — UNTESTED

A claim enters the substrate at UNTESTED status. UNTESTED claims may be surfaced in the cooperative review queue but carry no epistemic authority in the substrate’s KNOWN field. UNTESTED claims that fail to advance within a

configurable dormancy period (default: 90 days) are flagged for human review or archived.

Tier 2 — TESTED

A claim advances from UNTESTED to TESTED upon accumulating $K = 20$ independent verification attempts without a successful contradiction. Independent verification attempts may be contributed by: (a) automated cooperative plow pipeline runs (Claim Group 1 Loops 10 and 11 from Claim Group 28); or (b) cooperative member submissions. The K threshold is configurable per domain (default: $K = 20$ across all domains).

Tier 3 — FORGED

A claim advances from TESTED to FORGED upon either: - (a) surviving $M = 10$ adversarial contradiction attempts from Code Breaker members — attempts that specifically attempt to falsify the claim, not merely confirm it — without a successful falsification; or - (b) surviving without successful falsification for a configurable sustained period (default: 90 days from TESTED status).

The FORGED designation indicates that the claim has been subjected to intentional adversarial attack and has withstood it. FORGED claims carry elevated epistemic authority in the substrate’s KNOWN field and receive a provenance-premium in the Excalibur Active Currents pheromone decay scoring (Claim Group 14), decaying more slowly than non-FORGED claims of equivalent age and access frequency.

Tier 4 — GOLD_REFINED_BY_FIRE (Immutable)

A claim achieves GOLD_REFINED_BY_FIRE status — the Immutable designation — when: (a) the claim has achieved FORGED status; (b) no successful contradiction has been recorded in any subsequent challenge cycle; and (c) the cooperative’s designated Code Breaker verification committee (or the full Code Breaker membership, per cooperative bylaw) has reviewed the claim’s challenge history and confirmed the Immutable designation.

GOLD_REFINED_BY_FIRE claims are displayed in cooperative dashboards with a distinct visual indicator. They carry maximum epistemic authority in the substrate KNOWN field and are exempt from automated re-evaluation by Loops 10, 11, and 12 unless a reversal petition is filed (see Section 4 below).

The four-tier progression is summarized in the following table:

Tier	Name	Advancement Condition	Epistemic Authority
1	UNTESTED	Submission	Minimal
2	TESTED	$K=20$ independent verifications, no successful contradiction	Moderate
3	FORGED	$M=10$ adversarial Code Breaker challenges survived OR 90 days sustained	High
4	GOLD_REFINED_BY_FIRE	FORGED + all subsequent challenges survived +	Maximum (Immutable)

Tier	Name	Advancement Condition	Epistemic Authority
		committee review	

2. Code Breaker Member Tier Progression

Code Breaker membership is an open cooperative role. Any member may take on Code Breaker assignments by selecting claims from the cooperative's Code Breaker queue — the surfaced candidates produced by Loop 11 (Claim Group 28, element (d)). Code Breaker members advance through four member tiers based on accumulated successful falsification work.

Member Tier 1 — Initiate

Awarded upon first successful claim falsification. A successful falsification is defined as: the Code Breaker submits counter-evidence for a THEORIES_OPEN-status claim; the counter-evidence achieves a contradiction confidence above the configurable threshold (default: 0.70) per Loop 11 scoring; and the claim is migrated to ELIMINATED status as a result. The Code Breaker who submitted the successful counter-evidence is recorded as the eliminator in the ELIMINATED field.

Member Tier 2 — Journeyman

Awarded upon accumulation of 10 successful claim falsifications (10 claims migrated to ELIMINATED status with the member as recorded eliminator). Journeyman Code Breakers are eligible to challenge TESTED-tier claims, contributing to FORGED tier advancement (or prevention).

Member Tier 3 — Master

Awarded upon: (a) accumulation of 100 successful claim falsifications; AND (b) 10 successful challenges against FORGED-tier claims, where “successful” means the FORGED claim was reversed to TESTED status as a result of the Master Code Breaker's counter-evidence. Master Code Breakers are eligible to review claims for GOLD_REFINED_BY_FIRE designation, serving on the verification committee.

Member Tier 4 — Refiner of Gold

The highest Code Breaker honor. Awarded to members who have made 10 good-faith adversarial challenges against claims that subsequently achieved GOLD_REFINED_BY_FIRE status — where “good-faith” is assessed by the cooperative's Code Breaker committee based on the quality, rigor, and originality of the challenge attempt, regardless of whether the challenge succeeded in falsifying the claim.

The Refiner of Gold designation is a deliberate inversion of the conventional recognition structure. In standard verification systems, the highest honor goes to the person who discovers a new truth. In the Code Breakers Guild, the highest honor goes to the person who tried hardest and most honestly to break a truth — and failed. Their failure is the direct cause of the claim earning its Immutable status. The cooperative names and honors that contribution.

Economic parity: All four Code Breaker member tiers earn cooperative contribution currency (Marks) for successful falsification work. The Marks earned for falsification work are denominated as negative-knowledge-denomination Marks (see Section 5 below). A member who devotes their cooperative career entirely to falsification work — never affirming, always attacking — earns at economic parity with a member who devotes their career entirely to affirmation work.

3. The Reversal of Immutable Process

GOLD_REFINED_BY_FIRE status is not permanent by definition — it is permanent by evidence. The cooperative's position is that Immutable status means “no counter-evidence has been found that survives Code Breaker scrutiny,” not “no counter-evidence could ever exist.” History of science provides examples of paradigms that appeared fully verified and were later revised upon discovery of genuinely new empirical data not available during the prior verification period. A reversal petition may be filed by any Code Breaker member of Journeyman tier or above. A successful reversal petition requires:

(a) New Empirical Data. The petition must present empirical evidence that was not available during any prior challenge cycle — specifically, evidence generated or published after the most recent GOLD_REFINED_BY_FIRE designation.

4. The Cooperative Pays Its Critics

— Game-Theoretic Analysis

Traditional peer review is unpaid, anonymous, and structurally confirmation-biased. These three characteristics produce predictable pathologies: insufficient reviewer effort (unpaid reviewers minimize review burden), missed failures (anonymous reviewers bear no reputational cost for poor review), and systematic underdetection of false positives (confirmation bias in reviewer selection and motivation). Bug bounty programs in software security have demonstrated that paying adversarial verifiers specifically for finding failures — not for confirming security — reliably surfaces failures that unpaid internal review misses. The incentive structure of bug bounty is: the higher the severity of the bug found, the larger the reward.

5. Negative-Knowledge Tokens

— The Cooperative's Falsification Economy

Within the cooperative, members earn Marks as the cooperative's contribution currency. Under the standard model, Marks are earned for affirmation work: publishing verified content, completing cooperative services, delivering work that passes quality gates. These are confirmation-denomination Marks. The Code Breakers Guild introduces a second denomination: negative-knowledge-denomination Marks.

Patent Claims — Claim Group 29

(Claims drafted from canon sources cited in Yoke BP084:

canon_code_breakers_guild_gold_refined_by_fire_elimination_marks_bp084.

Verbatim claim substance from canon; patent-prose polish per PROV_22 style guide. Counsel finalizes formal language for non-provisional.)

Claim 29.1 (Code Breakers + Gold Refined — Independent System Claim)

A computer-implemented cooperative knowledge verification system comprising:

- (a) an UNTESTED tier for newly submitted knowledge claims entering the cooperative substrate; - (b) a TESTED tier for claims advancing from UNTESTED upon accumulation of a configurable number K of independent verification attempts without a successful contradiction, wherein K is configurable per domain and defaults to 20; - (c) a FORGED tier for claims advancing from TESTED upon either: (i) surviving a configurable number M of adversarial contradiction attempts from cooperative Code Breaker members without a successful falsification, wherein M defaults to 10; or (ii) surviving without successful falsification for a configurable sustained period following TESTED promotion, wherein the default period is 90 days; and - (d) a GOLD_REFINED_BY_FIRE tier, designating a claim as Immutable, awarded when a FORGED-tier claim has survived all subsequent challenge cycles and has been confirmed by a cooperative Code Breaker verification committee review; wherein each tier is mutually exclusive, each claim occupies exactly one tier at any moment, and advancement through tiers is monotonically increasing except upon admission of a reversal petition.

Claim 29.2 (Code Breakers + Gold Refined — Dependent)

The system of 29.1 further comprising a cooperative Code Breaker member tier progression comprising: - (a) an Initiate rank awarded upon first successful claim falsification, wherein a successful falsification is defined as submission of counter-evidence that achieves a contradiction confidence above a configurable threshold and causes a claim to migrate from THEORIES_OPEN status to ELIMINATED status; - (b) a Journeyman rank awarded upon accumulation of 10 successful claim falsifications; - (c) a Master rank awarded upon accumulation of 100 successful claim falsifications and 10 successful challenges against FORGED-tier claims causing those claims to revert from FORGED to TESTED status; and - (d) a Refiner of Gold rank awarded to members who have made a configurable number of good-faith adversarial challenges against claims that subsequently achieved GOLD_REFINED_BY_FIRE status, wherein the Refiner of Gold designation honors the member's role in the adversarial challenge process that earned the claim its Immutable designation, regardless of whether any individual challenge succeeded in falsification.

Claim 29.3 (Code Breakers + Gold Refined — Dependent: Independent System Claim)

A computer-implemented cooperative economic mechanism comprising: - (a) a cooperative contribution currency denominated as Marks; - (b) a negative-knowledge denomination tag applied at the ledger level to Marks earned specifically through successful demonstration that a knowledge claim is false, incomplete, or overstated, wherein the negative-knowledge denomination tag does not alter the Marks currency identity but records the epistemic nature of the work that produced the Marks; - (c) a redemption multiplier of 1.0x applied to negative-knowledge-denomination Marks, establishing economic parity between falsification-attributed Marks and affirmation-attributed Marks; - (d) a per-member cooperative dashboard metric surfacing the ratio of confirmation-denomination Marks to negative-knowledge-denomination Marks in each member's accumulated balance; and - (e) an aggregate cooperative dashboard metric surfacing the collective ratio of confirmation-denomination to negative-knowledge-denomination Marks across all active Code Breaker members.

Diagrams

Diagram E — Four-Tier Progression Flowchart (Figure 24)

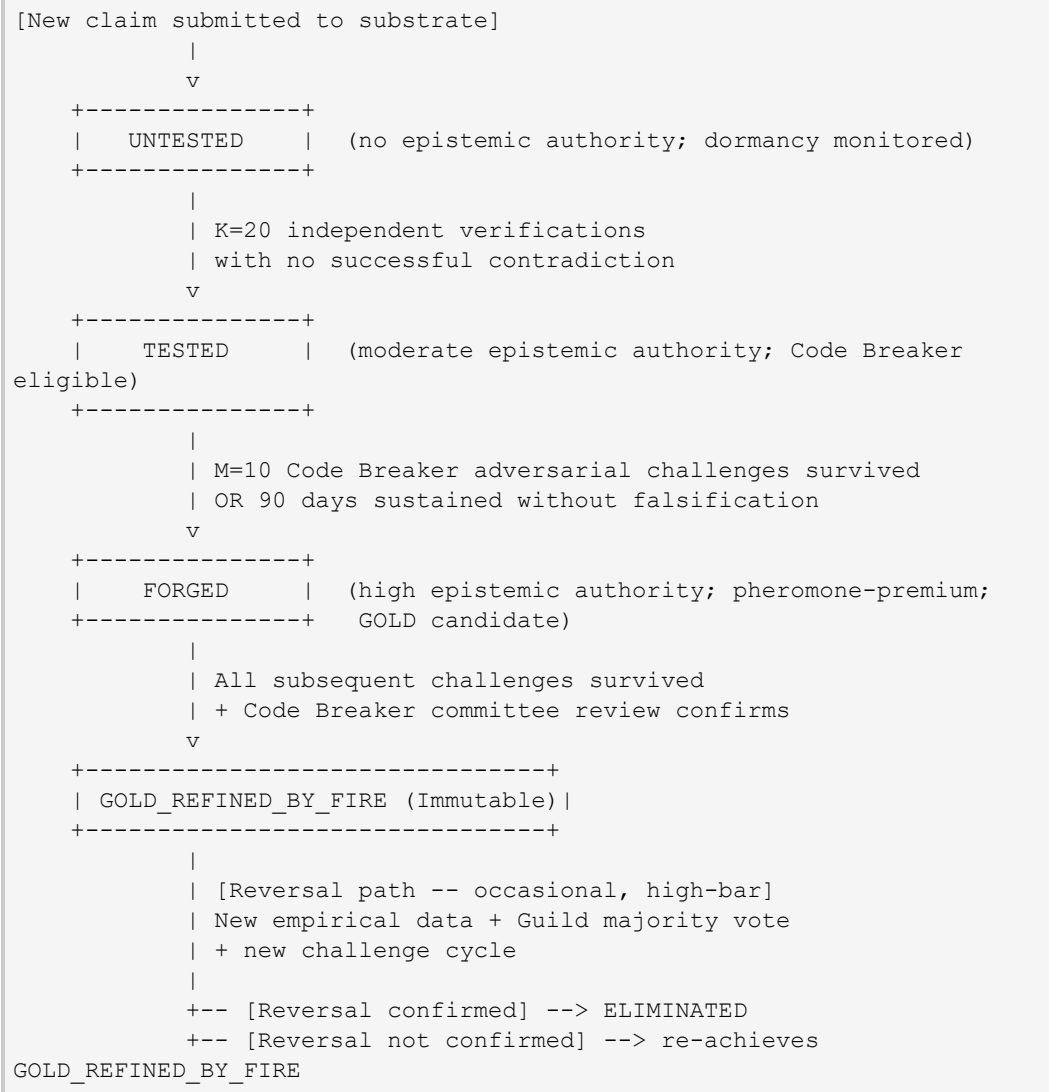


Figure 24: Four-tier knowledge claim progression — UNTESTED → TESTED → FORGED → GOLD_REFINED_BY_FIRE (Immutable), with configurable thresholds $K=20$, $M=10$, 90 days, and the reversal path requiring new empirical data and Guild majority vote.

Diagram F — Code Breaker Member Tier Progression (Figure 25)

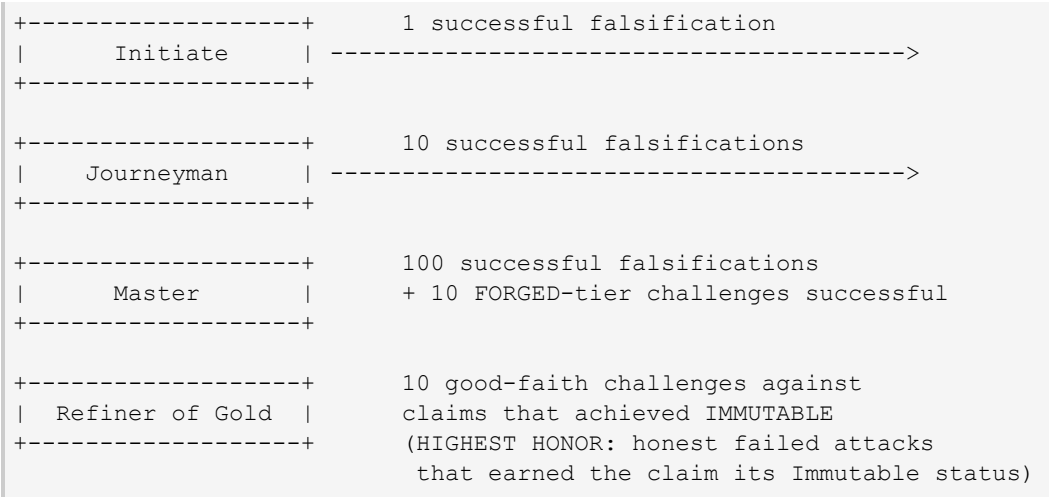


Figure 25: Code Breaker member tier progression — Initiate, Journeyman, Master, Refiner of Gold. The Refiner of Gold rank inverts conventional honor structure: highest rank awarded for honest hard attacks that FAILED, because those failed attacks directly caused claims to achieve Immutable status.

Diagram G — Negative-Knowledge Tokens Dashboard (Figure 26)

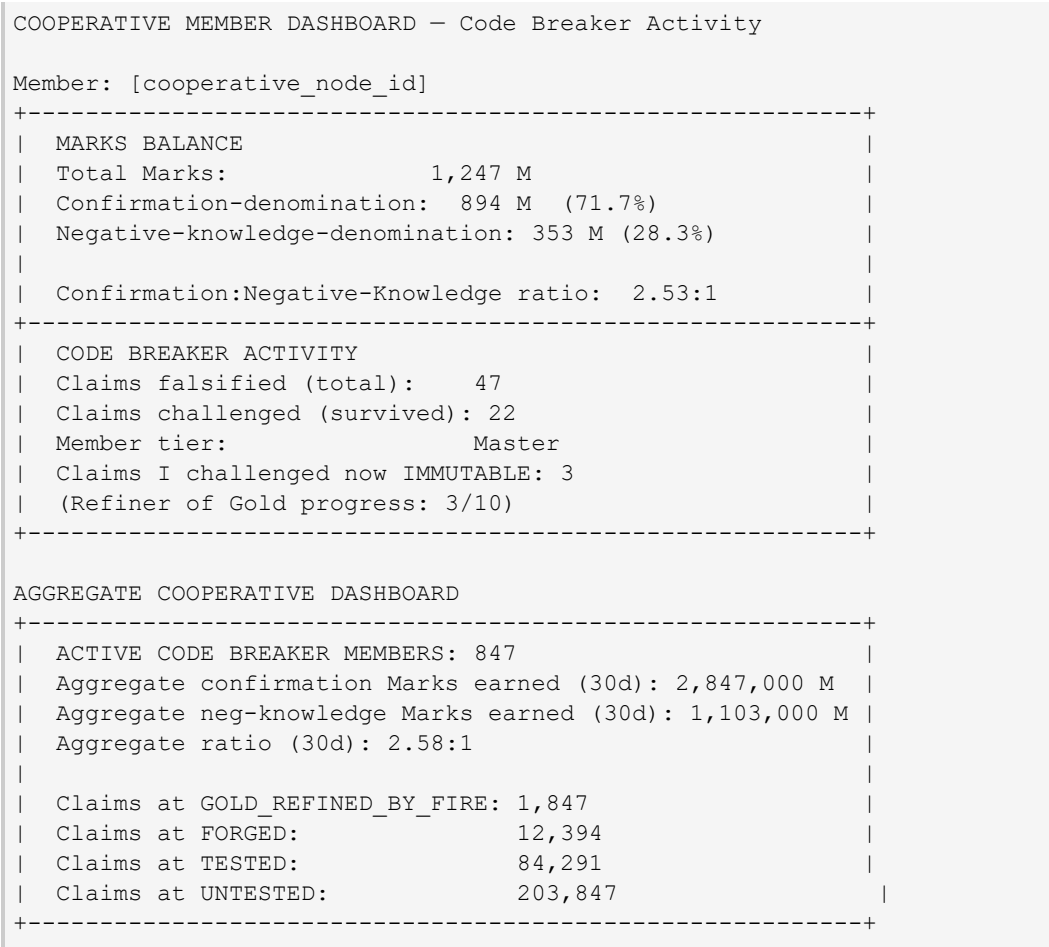


Figure 26: Negative-Knowledge Tokens dashboard — per-member view (confirmation vs. negative-knowledge Marks balance and ratio; Code Breaker tier progress; Refiner of Gold progress) and aggregate cooperative view (collective ratio; claim tier distribution).

Empirical Receipt and Reduction to Practice

The Code Breakers Guild and Gold Refined by Fire mechanism was ratified during BP084 session by Founder direct. The four-tier claim progression (UNTESTED → TESTED → FORGED → GOLD_REFINED_BY_FIRE) with thresholds K=20, M=10, and 90-day sustained period was ratified by Founder as the canonical verification ladder. The Refiner of Gold honor — awarding the highest Code Breaker distinction to members whose honest hardest attacks against eventually-Immutable claims failed — was ratified by Founder as a deliberate inversion of conventional recognition structure.

The negative-knowledge-denomination Marks mechanism was ratified as a tagged denomination within the existing Marks currency system, not a new cooperative currency. The 3-currency canon (Marks / Joules / Credits) is

preserved; negative-knowledge Marks are Marks with a denomination tag, not a fourth currency. The 1.0x redemption multiplier was ratified by Founder as the canonical economic parity signal.

The game-theoretic analysis (paying Code Breakers inverts peer-review perverse incentives; comparison to bug bounty prior art) is grounded in the published literature: Ioannidis (2005) on peer review pathology; bug bounty program empirical results across the software security industry from 2010-present.

This mechanism composes with and extends: Claim Group 1 (Canonical Plow Pipeline — Code Breaker queue is surfaced by Loop 11); Claim Group 12 (Substitution Rail Triple-Mode Exchange — Marks earned by Code Breakers circulate via the Marks rail, not fiat); Claim Group 14 (EAC Pheromone Decay — FORGED and GOLD_REFINED_BY_FIRE claims receive provenance-premium in decay scoring); Claim Group 26 (Substitution Rail Cooperative Marketplace — Medallion provenance stamps interact with claim tier); and Claim Group 28 (Truth Integrity Chain — Loop 11 surfaces Code Breaker candidates; Loop 10 survival scores inform tier progression).

Reduction to practice: the Code Breakers Guild queue is implemented in the 12-Blade Column Plow runtime as a structured output of Loop 11 (Claim Group 28, element (d)). The tier-progression logic is implemented as a configurable threshold checker on the eblet-level K and M counters. The negative-knowledge-denomination Marks ledger tag is implemented in the cooperative marks-ledger schema as an enum field on the Marks transaction record.

Cross-reference: Innovation Area 35 (Claim Group 28) — Truth Integrity Chain — defines the five-field TIC record (KNOWN / THEORIES_OPEN / ELIMINATED / DEPENDENCIES_UPSTREAM / APPLICATIONS_DOWNSTREAM) that Code Breakers operate upon. Innovation Area 35 must be read alongside Innovation Area 36 for full understanding of the verification architecture.

*Liana Banyan Corporation Inventor: J. Jones * Provisional Patent Application BP084 * Sonnet 4.6 * June 16, 2026 Innovation Area 36 * Claim Group 29 * status: founder-ratify-pending* — Claim Group 29*

(Claims drafted from canon sources cited in Yoke BP084: canon_code_breakers_guild_gold_refined_by_fire_elimination_marks_bp084. Verbatim claim substance from canon; patent-prose polish per PROV_22 style guide. Counsel finalizes formal language for non-provisional.)

Claim 29.1 (Code Breakers + Gold Refined — Independent System Claim)

A computer-implemented cooperative knowledge verification system comprising:

- (a) an UNTESTED tier for newly submitted knowledge claims entering the cooperative substrate;
- (b) a TESTED tier for claims advancing from

UNTESTED upon accumulation of a configurable number K of independent verification attempts without a successful contradiction, wherein K is configurable per domain and defaults to 20; - (c) a FORGED tier for claims advancing from TESTED upon either: (i) surviving a configurable number M of adversarial contradiction attempts from cooperative Code Breaker members without a successful falsification, wherein M defaults to 10; or (ii) surviving without successful falsification for a configurable sustained period following TESTED promotion, wherein the default period is 90 days; and - (d) a GOLD_REFINED_BY_FIRE tier, designating a claim as Immutable, awarded when a FORGED-tier claim has survived all subsequent challenge cycles and has been confirmed by a cooperative Code Breaker verification committee review; wherein each tier is mutually exclusive, each claim occupies exactly one tier at any moment, and advancement through tiers is monotonically increasing except upon admission of a reversal petition.

Claim 29.2 (Code Breakers + Gold Refined — Dependent)

The system of 29.1 further comprising a cooperative Code Breaker member tier progression comprising: - (a) an Initiate rank awarded upon first successful claim falsification, wherein a successful falsification is defined as submission of counter-evidence that achieves a contradiction confidence above a configurable threshold and causes a claim to migrate from THEORIES_OPEN status to ELIMINATED status; - (b) a Journeyman rank awarded upon accumulation of 10 successful claim falsifications; - (c) a Master rank awarded upon accumulation of 100 successful claim falsifications and 10 successful challenges against FORGED-tier claims causing those claims to revert from FORGED to TESTED status; and - (d) a Refiner of Gold rank awarded to members who have made a configurable number of good-faith adversarial challenges against claims that subsequently achieved GOLD_REFINED_BY_FIRE status, wherein the Refiner of Gold designation honors the member's role in the adversarial challenge process that earned the claim its Immutable designation, regardless of whether any individual challenge succeeded in falsification.

Claim 29.3 (Code Breakers + Gold Refined — Dependent: Independent System Claim)

A computer-implemented cooperative economic mechanism comprising: - (a) a cooperative contribution currency denominated as Marks; - (b) a negative-knowledge denomination tag applied at the ledger level to Marks earned specifically through successful demonstration that a knowledge claim is false, incomplete, or overstated, wherein the negative-knowledge denomination tag does not alter the Marks currency identity but records the epistemic nature of the work that produced the Marks; - (c) a redemption multiplier of 1.0x applied to negative-knowledge-denomination Marks, establishing economic parity between falsification-attributed Marks and affirmation-attributed Marks; - (d) a per-member cooperative dashboard metric surfacing the ratio of confirmation-

denomination Marks to negative-knowledge-denomination Marks in each member’s accumulated balance; and - (e) an aggregate cooperative dashboard metric surfacing the collective ratio of confirmation-denomination to negative-knowledge-denomination Marks across all active Code Breaker members.

Diagrams

Diagram E — Four-Tier Progression Flowchart (Figure 24)

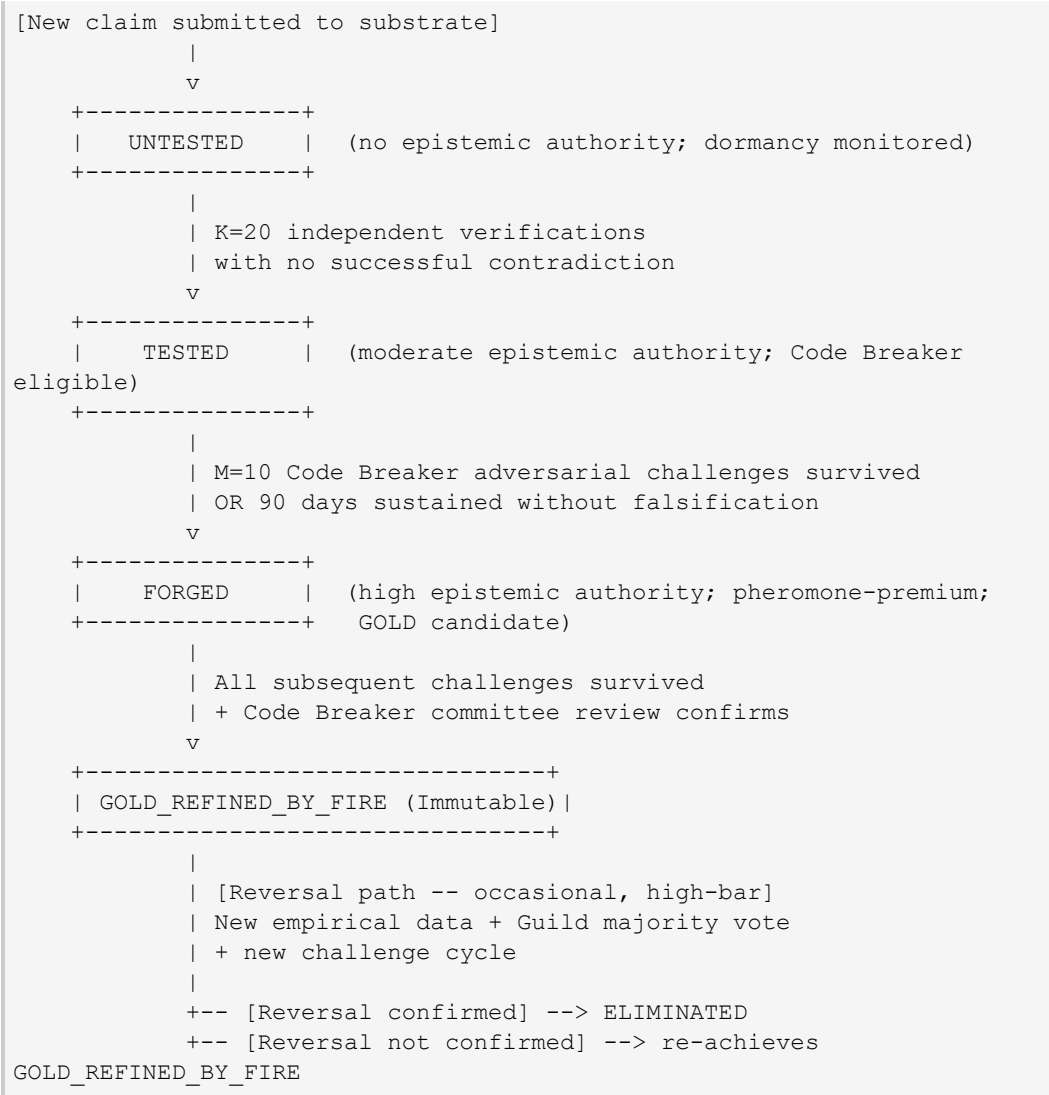


Figure 24: Four-tier knowledge claim progression — UNTESTED → TESTED → FORGED → GOLD_REFINED_BY_FIRE (Immutable), with configurable thresholds K=20, M=10, 90 days, and the reversal path requiring new empirical data and Guild majority vote.

Diagram F — Code Breaker Member Tier Progression (Figure 25)

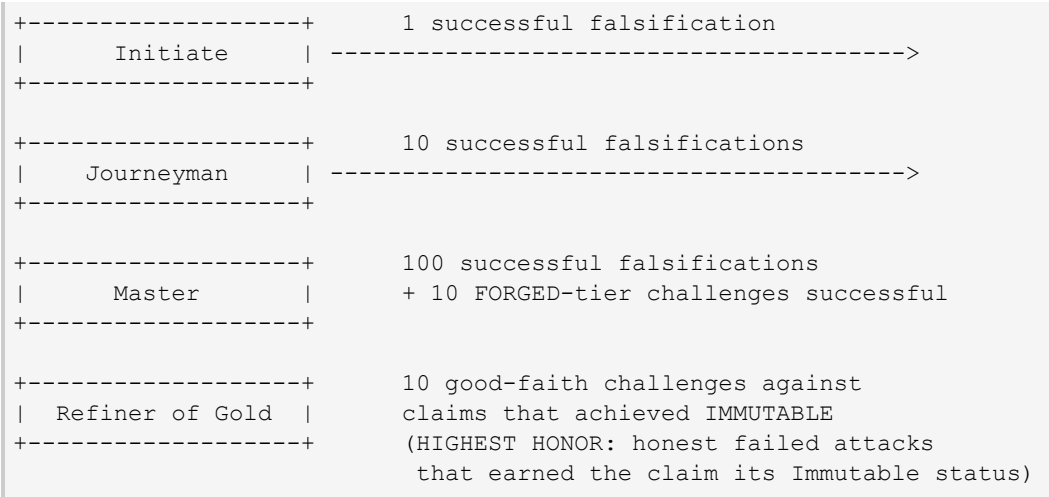


Figure 25: Code Breaker member tier progression — Initiate, Journeyman, Master, Refiner of Gold. The Refiner of Gold rank inverts conventional honor structure: highest rank awarded for honest hard attacks that FAILED, because those failed attacks directly caused claims to achieve Immutable status.

Diagram G — Negative-Knowledge Tokens Dashboard (Figure 26)

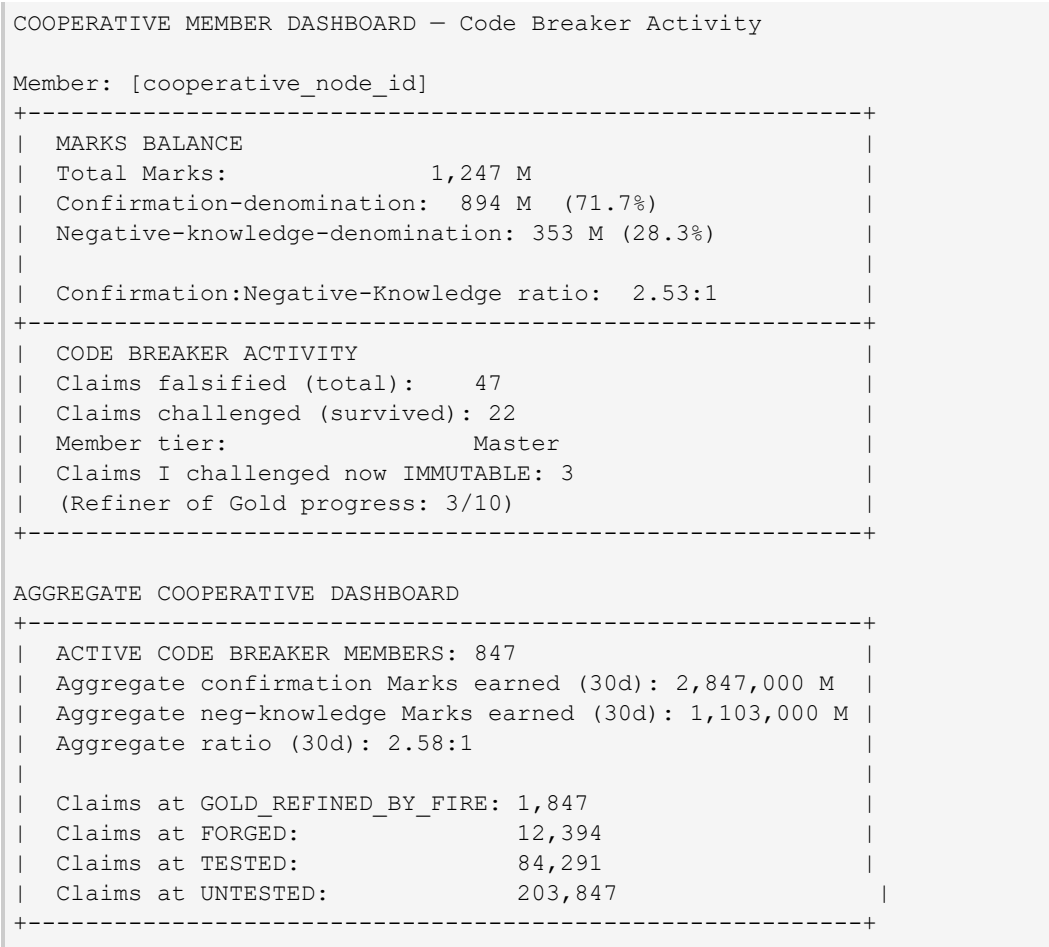


Figure 26: Negative-Knowledge Tokens dashboard — per-member view (confirmation vs. negative-knowledge Marks balance and ratio; Code Breaker tier progress; Refiner of Gold progress) and aggregate cooperative view (collective ratio; claim tier distribution).

Empirical Receipt and Reduction to Practice

The Code Breakers Guild and Gold Refined by Fire mechanism was ratified during BP084 session by Founder direct. The four-tier claim progression (UNTESTED → TESTED → FORGED → GOLD_REFINED_BY_FIRE) with thresholds K=20, M=10, and 90-day sustained period was ratified by Founder as the canonical verification ladder. The Refiner of Gold honor — awarding the highest Code Breaker distinction to members whose honest hardest attacks against eventually-Immutable claims failed — was ratified by Founder as a deliberate inversion of conventional recognition structure.

The negative-knowledge-denomination Marks mechanism was ratified as a tagged denomination within the existing Marks currency system, not a new cooperative currency. The 3-currency canon (Marks / Joules / Credits) is

preserved; negative-knowledge Marks are Marks with a denomination tag, not a fourth currency. The 1.0x redemption multiplier was ratified by Founder as the canonical economic parity signal.

The game-theoretic analysis (paying Code Breakers inverts peer-review perverse incentives; comparison to bug bounty prior art) is grounded in the published literature: Ioannidis (2005) on peer review pathology; bug bounty program empirical results across the software security industry from 2010-present.

This mechanism composes with and extends: Claim Group 1 (Canonical Plow Pipeline — Code Breaker queue is surfaced by Loop 11); Claim Group 12 (Substitution Rail Triple-Mode Exchange — Marks earned by Code Breakers circulate via the Marks rail, not fiat); Claim Group 14 (EAC Pheromone Decay — FORGED and GOLD_REFINED_BY_FIRE claims receive provenance-premium in decay scoring); Claim Group 26 (Substitution Rail Cooperative Marketplace — Medallion provenance stamps interact with claim tier); and Claim Group 28 (Truth Integrity Chain — Loop 11 surfaces Code Breaker candidates; Loop 10 survival scores inform tier progression).

Reduction to practice: the Code Breakers Guild queue is implemented in the 12-Blade Column Plow runtime as a structured output of Loop 11 (Claim Group 28, element (d)). The tier-progression logic is implemented as a configurable threshold checker on the eblet-level K and M counters. The negative-knowledge-denomination Marks ledger tag is implemented in the cooperative marks-ledger schema as an enum field on the Marks transaction record.

Cross-reference: Innovation Area 35 (Claim Group 28) — Truth Integrity Chain — defines the five-field TIC record (KNOWN / THEORIES_OPEN / ELIMINATED / DEPENDENCIES_UPSTREAM / APPLICATIONS_DOWNSTREAM) that Code Breakers operate upon. Innovation Area 35 must be read alongside Innovation Area 36 for full understanding of the verification architecture.

Claim 29.4 (Code Breakers + Gold Refined — Dependent)

The system of 29.1 wherein the FORGED tier designation composes with the Excalibur Active Currents pheromone decay mechanism (Claim Group 14) such that FORGED-tier claims receive a provenance-premium in EAC decay scoring, decaying more slowly than non-FORGED claims of equivalent age and access frequency, reflecting the higher epistemic value represented by surviving adversarial Code Breaker challenges.

Claim 29.5 (Code Breakers + Gold Refined — Dependent)

The system of 29.1 wherein the Code Breaker assignment queue is populated by the Elimination Verification loop (Loop 11) of the Truth Integrity Chain mechanism (Claim Group 28) upon each elimination event, linking the Code Breaker verification system to the automated hypothesis-elimination pipeline

such that AI-pipeline-detected elimination candidates are automatically surfaced for human Code Breaker verification before permanent ELIMINATED assignment.

Empirical Receipt

Code Breakers ratified BP084. Ioannidis 2005 replication crisis: unpaid reviewers lack adversarial incentive. Bug bounty programs: closest prior art. Refiner of Gold honor (inverted reward for failed attacks on eventually-Immutable claims) resolves confirmation-bias structurally.

Claim Group 30: Unseen Tax — Real-Time AI Cost Transparency

Anecdote — Founder Voice

Founder verbatim — the recognition: > “Remember the paper on ‘Unseen Tax’ and the feature that we offer that doesn’t hide your ai subscriptions cost with the whole ‘default to Composer 2.5’ - the MOST expensive credit to use, etc.?”

— Founder, BP084 (verbatim)

Founder verbatim — the ratify: > “YES mint the unseen tax canon”

— Founder, BP084 (verbatim)

The name is precise. A tax is a charge collected by an authority from a party who has no practical ability to refuse and often no clear visibility into the amount being extracted. That is what happens every time a developer opens an AI coding tool and begins working.

Cursor defaults to Composer 2.5 — the most expensive credit in the tool’s pricing tier. Many AI agents default to premium model tiers first. The model selection is not surfaced to the user at the moment of the call. No per-call cost is shown while the work is being done. At the end of the session, the user sees credits drained. They cannot reconstruct which calls produced leverage and which were waste. The billing surface offered is aggregate-only — a monthly total with no itemization by call, by purpose, or by session. There is no comparison surface: no notification that this prompt cost \$0.42 via Composer 2.5 and would have cost \$0.03 via a capable smaller model.

The industry has converged on five structural behaviors that constitute the Unseen Tax:

1. **Default to the most expensive credit** — Cursor → Composer 2.5; many AI agents → premium-tier model first, without disclosure
2. **Hide per-call cost** during the session — no live ticker, no per-message marginal cost surface while the work is being done
3. **Surface only the aggregate at month-end** — by which point the user cannot disentangle which calls were leverage and which were waste
- 4.

Background and Problem Statement

The market for artificial intelligence developer tooling — including but not limited to automated code completion, multi-step autonomous agent pipelines, inference-backed documentation assistants, and natural language query interfaces — has converged on a billing architecture that structurally conceals the per-unit cost of inference from the end user at the time of consumption. This architecture is not accidental. It serves the economic interest of the platform provider: users who cannot see the per-call cost of model inference cannot make substitution decisions. Users who cannot make substitution decisions continue to consume the most expensive available model tier regardless of task-class requirements. The resulting aggregate revenue per user is structurally higher than it would be under full per-call cost transparency.

No prior AI developer tool in wide deployment provides: (a) real-time per-call marginal cost surfacing at the moment of inference, displayed to the user before or coincident with the result; (b) a continuously visible running session total updated after each inference call; (c) a comparison delta annotated at the moment of model escalation, disclosing to the user both the cost differential and the machine-generated reason for escalation; (d) a system default that selects the lowest-priced model meeting a configurable task-class threshold, requiring explicit user override or demonstrated capability insufficiency before escalating to a higher-priced model; (e) an itemized post-session receipt categorizing each inference call by purpose, model tier, and cost; or (f) a cross-vendor monthly aggregate view that unifies spend across multiple distinct artificial intelligence subscription services into a single member-facing dashboard organized by vendor, purpose, and user-defined category.

The Unseen Tax innovation area addresses this gap by specifying a cooperative-class cost-transparency architecture that treats per-call inference cost as a first-class data type required to be surfaced to the member at the time of consumption — composing with the Personal Subscription Steward tier architecture, the Code Breakers Guild transparency commitment, and the Truth-Always BLOOD discipline applied at the financial layer.

Detailed Description**1. Real-Time Per-Call Marginal Cost Surface**

The cooperative-class application maintains a connection to the vendor inference API's pricing schedule for each model tier available to the application. At each inference call, the application resolves the cost of that specific call — calculated from: (a) input token count; (b) output token count; (c) model tier selected; and (d) current vendor pricing — and surfaces the marginal cost to the member in real time, coincident with the delivery of the inference result.

The per-call cost surface is not deferred to end-of-session. It is not deferred to a settings or billing menu. It is displayed at the point of interaction, in the same interface layer where the inference result is presented. A member reviewing the result of an inference call sees, in the same view: the result, and what it cost.

The cost surface includes: the marginal cost of this specific call; the model tier used; and whether this call was handled by the default cheapest-capable model or an escalated tier.

2. Running Session Total

The application maintains a session-level cost accumulator initialized to zero at the start of each member session. After each inference call, the accumulator is incremented by the marginal cost of that call. The running session total is visible in the application's primary interface throughout the session — not in a submenu, not behind a settings toggle, not in a billing panel accessible only after session end.

The running session total is updated synchronously with each call's cost surface. A member who has consumed seven inference calls in a session can, at any point before the eighth call, see exactly what the first seven calls have cost.

3. Comparison Delta on Escalation

When the application determines that a task-class requirement cannot be satisfied by the default cheapest-capable model and escalates to a higher-priced model tier, the following information is surfaced to the member at the moment of escalation:

- a. **cost-this-call**: the marginal cost of the call at the escalated tier;
- b. **cost-at-default**: the marginal cost the call would have incurred at the default cheapest-capable tier;
- c. **delta**: the cost differential between escalated and default, expressed as an absolute amount and as a percentage premium;
- d. **escalation-reason**: a machine-generated natural-language annotation stating the specific capability insufficiency that caused the escalation (e.g., “escalated from [model-A] to [model-B] because task-class requires extended context window exceeding [model-A] maximum”).

The comparison delta is not an optional disclosure. It is a structural requirement of the escalation flow: no escalation executes without surfacing (a) through (d) to the member.

4. Default to Cheapest Available

The application's default model selection algorithm selects the lowest-priced available model meeting a configurable task-class threshold. Task-class thresholds are configurable by: (a) the application operator; (b) the member, within operator-defined bounds; and (c) a per-task-type classification system that

assigns each inference request to a task class (e.g., code completion, natural language query, multi-step agentic pipeline, document summarization) with corresponding minimum capability requirements.

Escalation from the default cheapest-capable model to a higher-priced tier requires either: (a) explicit member override, confirmed at the time of the request; or (b) demonstrated insufficient capability — defined as an empirically measured failure of the default model to meet the task-class threshold on this specific request, with the failure evidence surfaced to the member.

The application is prohibited from defaulting to the most expensive available model tier by structural enforcement: the model selection algorithm requires an explicit reason — either override or demonstrated insufficiency — to depart from the cheapest-capable default.

5. Itemized Post-Session Receipt

At the close of each member session, the application generates an itemized receipt comprising one line item per inference call, each containing: (a) call sequence number within session; (b) timestamp; (c) task class; (d) model tier used; (e) marginal cost; (f) whether the call was default or escalated; and (g) if escalated, the escalation reason and cost delta.

The itemized post-session receipt is stored in the member's cooperative account and accessible at any subsequent time. It is not discarded at session close. The receipt constitutes the permanent record of what inference was consumed, for what purpose, at what cost — composing with the cooperative-class eblet substrate's append-only audit trail architecture.

6. Cross-Vendor Aggregate — Personal Subscription Steward Composition

The Personal Subscription Steward (per `canon-mnemosynec-personal-subscription-steward-tier-ramp-bp084`) extends the per-session cost surface to a cross-vendor monthly aggregate view. The member's monthly spend across multiple distinct AI subscription providers — including but not limited to Anthropic Claude, Cursor, Google Gemini, Perplexity, ChatGPT, and other tracked providers — is unified into a single cooperative-member-facing dashboard.

The cross-vendor aggregate view is organized by: - (a) **vendor** — monthly spend per subscription provider, ranked by spend; - (b) **purpose** — spend categorized by task class across all vendors (e.g., total monthly spend on code completion across all tools; total monthly spend on document query); - (c) **user-defined category** — member-configurable spending categories enabling personalized budget tracking (e.g., “client work,” “personal projects,” “open source contributions”).

The cross-vendor aggregate surfaces: (i) a running monthly total visible at every member interaction; (ii) per-vendor breakdowns with trend lines; (iii) substitution recommendations when spend concentration in a high-cost vendor for a task class

could be reduced by substituting a lower-cost vendor with equivalent capability for that class.

The Personal Subscription Steward composition enables the member to answer, at any moment: *“How much have I spent on AI this month, across all tools, and what did each dollar accomplish?”*

7. The Six Hard-Banned Behaviors

The cooperative-class architecture structurally prohibits the following behaviors, which collectively constitute the Unseen Tax anti-pattern. These prohibitions are enforced by the application’s architecture, not merely by policy:

Hard-Ban 1 — default-to-expensive: The application may not default to the most expensive available model tier without surfacing the cost and the reason. The default must be the cheapest capable tier.

Hard-Ban 2 — hide-per-call: The application may not conceal the per-call marginal cost of an inference call. The cost is surfaced coincident with the result, not deferred.

Hard-Ban 3 — aggregate-only-billing: The application may not provide only an aggregate billing surface (monthly total, credit balance) without itemized per-call breakdown available. The itemized receipt is always accessible.

Hard-Ban 4 — escalate-without-delta: The application may not escalate to a higher-priced model tier without surfacing the cost delta and escalation reason at the moment of escalation.

Hard-Ban 5 — bury-in-settings: The application may not require the member to navigate to a settings, advanced, or billing submenu to access per-call cost data, running session total, or itemized receipts. These are primary-interface elements.

Hard-Ban 6 — cost-obfuscation-marketing: The application and its marketing copy may not use cost-obfuscating framing — including “unlimited!” when credit caps exist, “all-you-can-use!” when throttles apply, or aggregate pricing language that suppresses per-unit cost visibility. Every marketing representation of cost is compatible with the per-call cost surface architecture.

(Formal independent and dependent claims for Claim Group 30. Language preserved verbatim from canon ratification per canon-unseen-tax-anti-pattern-ai-cost-transparency-bp084. Formal independent and dependent claims will be crystallized in the corresponding non-provisional application.)

Claim 30.1 — A computer-implemented method for AI inference cost transparency in a member-class application, wherein each call to a vendor inference API surfaces its marginal cost to the user in real time, displays a running session total, and exposes a comparison delta when the application escalates from a cheaper model tier to a more expensive one, said delta annotated with the reason for escalation.

Claim 30.2 — The method of claim 30.1, further comprising an aggregate-across-vendors view in which the application tracks the user's monthly spend across multiple distinct AI subscription providers and surfaces the breakdown by vendor, by purpose, and by user-defined category, enabling the user to make informed substitution decisions.

Claim 30.3 — The method of claim 30.1, wherein the application's default model selection is the lowest-priced available model meeting a configurable task-class threshold, with escalation to higher-priced models requiring either explicit user override or demonstrated insufficient capability via empirical evidence.

Claim 30.4 The method of claim 30.1, wherein the six Hard-Banned behaviors (default-to-expensive, hide-per-call, aggregate-only-billing, escalate-without-delta, bury-in-settings, cost-obfuscation-marketing) are enforced by the application's architecture such that the model selection algorithm, billing display layer, and marketing copy generation are each structurally constrained from exhibiting any Hard-Banned behavior, and any application update introducing a Hard-Banned behavior fails an automated structural compliance check before deployment.

Claim 30.5 The method of claim 30.1, wherein the itemized post-session receipt is stored as a cooperative-class eblet in the member's local substrate store using the append-only audit trail architecture, each receipt record comprising call sequence number, timestamp, task class, model tier, marginal cost, escalation flag and reason, and a SHA-256 hash of the receipt content enabling tamper detection.

Empirical Receipt

Unseen Tax ratified BP084 Founder verbatim: "YES mint the unseen tax canon." Originating instance: Cursor default-to-Composer-2.5. Phase 1 MnemosyneC metrics receipt (session cost in yoke-receipts) constitutes first primitive instance. Six Hard-Banned behaviors constitute novel AI billing anti-pattern classification.

Claim Group 31: MnemosyneC-as-Interface — Vendor-Resilient Persistent AI Host

31.1 (independent) — Vendor-Resilient Persistent AI Host Architecture

A system for persistent-host vendor-resilient artificial intelligence interface, comprising: a local application instance constituting a user's primary AI workspace; a local substrate store of vendor-agnostic knowledge records persisting user context, domain assignments, history, and preferences independent of any specific AI vendor; a vendor health monitor that polls registered AI vendor endpoints and maintains a current availability registry; a substrate worker routing layer that, upon receiving a user task, selects a healthy vendor from the availability registry, dispatches the task to the selected vendor via the substrate worker API, and records the task result in the local substrate

store; and a continuity guarantee such that workspace state persists across vendor unavailability events without requiring user re-initialization, migration, or context re-establishment. (Source:

canon_mnemosynec_as_interface_persistent_host_vendor_resilient_bp085.eb1 et.md; roadmap §1)

31.2 (dependent on 31.1) — Persistent Active Memory Crown Jewel

The system of Claim 31.1, wherein the local substrate store maintains a Persistent Active Memory record designated as the Crown Jewel, comprising: a current-session context summary updated after each task dispatch; a domain-assignment registry mapping subject domains to currently assigned AI substrate workers; a cross-vendor continuity record such that knowledge recorded during sessions using Vendor A is accessible and applicable during subsequent sessions using Vendor B; and a memory keyhole index enabling rapid context rehydration at session start without requiring vendor-side conversation history retrieval.

(Source: roadmap §1, Task 73 crown-jewel memory canon)

31.3 (dependent on 31.1) — Continuity-Lift Second-Order Value

The system of Claim 31.1, wherein each task dispatched through the routing layer produces a substrate knowledge record that compounds the user's accumulated domain knowledge independently of which vendor performed the task, such that: the user's effective AI capability increases monotonically with usage regardless of vendor substitution events; and the departure or unavailability of any single vendor does not reduce the user's accumulated substrate knowledge, creating a continuity-lift property wherein the persistent host application provides compounding second-order value exceeding the first-order value of any single vendor session. (Source: Substrate Theorem BP061 extension; roadmap §1 “she IS the continuous host”)

Claim 31.4 (MnemosyneC Persistent Host — Dependent Claim)

(dependent on 31.1) The system of 31.1 wherein the vendor health monitor maintains a priority-ordered vendor queue ranking vendors by a composite score comprising current per-token cost, recent response latency, and member-configured preference weight, such that task routing minimizes inference cost by default, composing with the Consult-don't-Rent task-class routing discipline of Claim Group 33.

Claim 31.5 (MnemosyneC Persistent Host — Dependent Claim)

(dependent on 31.1) The system of 31.1 wherein at session close the local substrate store executes a session-boundary re-weave operation condensing session-accumulated task results and knowledge records into persistent eplets, enabling the next session to initialize from the condensed substrate rather than from a raw session transcript, ensuring each session's knowledge compounds into the persistent substrate regardless of which vendor was used.

Claim Group 32: MedLab — Concoctions, Reins Assignment, and Comptroller Sub-Agent

32.1 (independent) — MedLab Executable AI-Fix Recipe Module

A system for eblet-stored executable AI-connection-fix recipes, comprising: a MedLab module within a persistent AI host application; a library of Concoction records, each Concoction comprising a machine-readable trigger condition defining the failure state that activates the recipe, a typed ingredient manifest classifying required inputs as configuration-value ingredients, log-evidence ingredients, or environment-variable-name ingredients, an ordered execution procedure defining discrete steps to resolve the failure condition, and a ratify gate defining an empirical success criterion that must be confirmed before the Concoction is marked as successfully applied; a Concoction browser interface enabling the host application to surface matching Concoctions when a failure condition is detected; and a Concoction execution audit log recording trigger event, applied Concoction identifier, ingredient values used, procedure steps completed, ratify gate outcome, and session timestamp. (Source:

canon_medlab_eblet_recipes_ai_connection_fixes_bp085.eblet.md;
canon_concoctions_medlab_ingredients_recipe_anatomy_bp085.eblet.md;
roadmap §4)

32.2 (dependent on 32.1) — Concoction Peer Contribution and Verification

The system of Claim 32.1, wherein Concoction records are contributed by cooperative members via a Package Store submission pathway, and wherein each contributed Concoction undergoes adversarial verification by Code Breaker Guild members per the GOLD_REFINED_BY_FIRE tier progression of Claim Group 29 before the Concoction is designated as trusted for automatic execution; such that the Concoction library compounds in accuracy and coverage as the cooperative membership grows, and members earn cooperative Marks denominations for contributing and adversarially verifying Concoctions. (Source: roadmap §4; composition with Code Breakers CG29)

32.3 (independent) — Reins Assignment Per-Category Dynamic AI Domain Registry

A method for dynamic artificial intelligence domain assignment in a persistent AI host application, comprising: maintaining a Reins Registry comprising records each associating a subject domain with an assigned AI substrate worker, a ratify gate defining criteria for valid assignment, and a timestamp of last assignment; upon initiating a task belonging to a registered domain, routing the task to the domain's currently assigned substrate worker rather than to a default worker; recording the outcome of each domain task as a substrate eblet associated with the domain assignment; and upon reassigning a domain to a new substrate worker, logging the prior holder, the reassignment timestamp, and a rationale note, such that domain expertise accumulated in prior assignments remains

accessible in the substrate store irrespective of which substrate worker currently holds the domain assignment. (Source:

`canon_mnemosynec_assigns_reins_per_category_bp085.eblet.md`; roadmap §6)

32.4 (dependent on 32.3) — Dedicated Sub-Agent One Role · Comptroller Pattern

The method of Claim 32.3, wherein a substrate worker is instantiated as a Dedicated Sub-Agent assigned exclusively to one persistent domain role, the Comptroller role comprising: periodic polling of an external financial catalog API to retrieve current product and pricing records; comparison of the retrieved records against a prior-state catalog eblet to produce a delta report identifying created, modified, or archived items; generation of an escalation notification upon detection of any delta; and dispatch of the delta report to the host application's session-start report queue; wherein the Comptroller sub-agent does not retain or transmit financial API credential values, references environment variable names only, and operates within the zero-credential-exposure discipline of the persistent host's secrets-handling canon. (Source:

`canon_dedicated_sub_agent_one_role_comptroller_pattern_bp085.eblet.md`; roadmap §7)

Claim 32.5 (MedLab / Concoctions / Reins — Dependent Claim)

(dependent on 32.1) The system of 32.1 wherein each Concoction record maintains a version history comprising the original text, subsequent amendments, the member identifier of each amending contributor, and a ratify-gate outcome record for each amendment, such that a superseded version may be restored if a subsequent amendment degrades diagnostic accuracy.

Claim 32.8 (MedLab / Concoctions / Reins — Dependent Claim)

(dependent on 32.3) The method of 32.3 wherein domain assignment conflicts are resolved by a configurable priority rule selecting among: the worker with the most recent successful domain task completion; the worker with the highest current health score; or the worker explicitly designated by the member; with conflict resolution recorded as a substrate event.

Claim 32.9 (MedLab / Concoctions / Reins — Dependent Claim)

(dependent on 32.3) The method of 32.3 wherein a Reins handoff ceremony transfers domain authority from one substrate worker to another, comprising: serialization of the outgoing worker's domain-specific context to a substrate eblet; transmission of the serialized context to the incoming worker; confirmation by the incoming worker that context has been received; and recording of the handoff event in the domain assignment log.

Claim Group 33: NetLinkWebNode — Consent-Based Peer AI-Connection Borrowing and Consult-Don't-Rent

33.1 (independent) — Peer AI-Connection Borrowing · NetLinkWebNode

A system for consent-based peer artificial intelligence connection lending and borrowing within a cooperative node mesh, comprising: a Constellation Switchboard maintaining peer node presence records including connection health state for each registered node; a connection-lending consent interface through which a node operator may designate their active AI vendor connections as available for cooperative peer borrowing; a borrowing request pathway through which a node lacking a healthy vendor connection may request temporary access to a consenting lender node's connection; a usage metering component recording connection-borrowing events with duration and task count attributed to the borrowing node; and a Marks settlement layer crediting the lending node and debiting the borrowing node in cooperative Marks denominations proportional to measured usage, such that AI vendor access is redistributed within the cooperative mesh according to member consent and settled through the cooperative's three-currency economy without requiring the lender to expose vendor API credential values to the borrower. (Source:

canon_netlinkwebnode_mic_vendor_resilient_peer_cluster_bp085.eblet.md; roadmap §3 v0.7.x)

33.2 (dependent on 33.1) — Mesh Interconnect Coordinator Election

Algorithm

The system of Claim 33.1, further comprising a Mesh Interconnect Coordinator (MIC) election algorithm that: among all nodes in the active state within the Constellation Switchboard, selects the node having the highest reputation-weighted Marks balance as the cluster coordinator; assigns the coordinator responsibility for routing peer connection-borrowing requests to available lending nodes; reelects a new coordinator upon coordinator node departure or state change to inactive; and records each coordinator election event as a substrate eblet with node identifier, election timestamp, reputation score at election, and predecessor coordinator identifier. (Source:

canon_netlinkwebnode_mic_vendor_resilient_peer_cluster_bp085.eblet.md; roadmap §3 v1.0)

33.3 (independent) — Consult-Don't-Rent Task-Class AI Cost Routing

A method for task-class-based artificial intelligence model selection in a persistent AI host application, comprising: maintaining a task-class routing table associating task categories with a minimum-capable model tier selected from at minimum a free-local-model tier and a premium-vendor tier; upon receiving a user task, classifying the task into a task class and selecting the minimum-capable tier from the routing table; dispatching the task to a model in the selected tier; recording a per-task cost receipt comprising the task class, model selected, and a cost delta expressing the difference between the selected model's cost and the next-tier premium model's cost; accumulating cost receipts into a running session total visible to the user throughout the session; and requiring a demonstrated-insufficiency event or explicit user override before escalating a task to a higher-

cost tier, wherein escalation generates an escalation-delta disclosure stating the incremental cost and machine-generated rationale, composing with the Unseen Tax anti-pattern disclosure of Claim Group 30. (Source:

canon_consult_ai_flagships_dont_rent_the_c40_aircraft_bp085.ebilet.md; roadmap §8)

Claim 33.4 (NetLinkWebNode / Consult-Dont-Rent — Dependent Claim)

(dependent on 33.1) The system of 33.1 wherein the usage metering component records borrowing events at configurable granularity (default: per-task), and the Marks settlement layer processes accumulated metering records at a configurable settlement interval (default: 24 hours), enabling settlement to be batched while maintaining per-task attribution accuracy.

Claim 33.7 (NetLinkWebNode / Consult-Dont-Rent — Dependent Claim)

(dependent on 33.3) The method of 33.3 wherein each inference call's cost receipt composes with the Unseen Tax Real-Time AI Cost Transparency mechanism (Claim Group 30) such that escalation events generated by the Consult-don't-Rent routing produce comparison-delta disclosures per the Unseen Tax architecture, and the running session cost total is the same total displayed by the Unseen Tax mechanism.

Claim Group 34: Many Doors One Cooperative — Multi-Entrance Unified Membership and Cue Deck Card

34.1 (independent) — Many Doors One Cooperative Multi-Entrance Membership Architecture

A system for multi-entrance unified cooperative membership, comprising: a cooperative membership platform defining a single canonical membership class with uniform rights including governance participation, cooperative economics, and substrate access; an entrance pathway registry storing an unbounded set of named entrance pathways, each pathway comprising a domain-specific value proposition, a contextually adapted membership framing, an optional inviting-member attribution identifier, and a journey-fork configuration defining the enrollment sequence presented to prospective members arriving at that pathway; a membership enrollment processor that, upon a prospective member completing enrollment through any registered pathway, creates a single canonical cooperative membership record irrespective of which entrance pathway was used; and an attribution ledger recording the entrance pathway identifier and inviting-member identifier associated with each enrolled member, wherein attribution persists permanently in the member record and qualifies the inviting member for cooperative Marks earnings as defined by the cooperative's attribution economy. (Source: roadmap §3 composition; Cue Deck Card system; Many Doors One Cooperative canon)

34.2 (dependent on 34.1) — Cue Deck Card Personalized Invitation System

The system of Claim 34.1, wherein each entrance pathway is associated with a Cue Deck Card record comprising: a recipient identifier or role designation; a personal message composed by the inviting member or cooperative administrator; a value-proposition narrative adapted to the recipient's domain context, geographic context, or demonstrated need; one or more journey-fork options presented to the recipient enabling selection of a cooperative initiative aligned with the recipient's context; a unique short-code or URL resolving to the entrance pathway; and a share configuration specifying language, sender attribution, and journey-fork set; such that a prospective member receiving a Cue Deck Card encounters a cooperative membership invitation that references their specific context, names their qualification as a community asset rather than a credential, and routes them to the most contextually relevant cooperative initiative. (Source: Cue Deck Card system `cue_card_system.sql`; Captain First Node Cue Card B107; roadmap §3)

34.3 (dependent on 34.1) — Captain's Ship Wheel Node Orchestrator Pattern

The system of Claim 34.1, wherein a node captain role is defined as a cooperative member who: manages one or more entrance pathways associated with a geographic node or domain-specific initiative; views a node dashboard displaying enrolled members, active journey forks, and attribution earnings; earns Captain Marks denominations for each member enrolled through pathways they manage; holds a permanent governance seat in the node's local governance upon the node reaching a defined membership threshold; and operates within the cooperative's unified bylaws without requiring a separate legal entity, cooperative charter, or fee structure, such that the Captain's authority is architecturally scoped to their node's entrance pathways while their members' cooperative rights are governed by the platform's unified membership class. (Source: roadmap §3; Captain First Node architecture B107; Many Doors One Cooperative canon)

Claim 34.4 (Many Doors / Cue Deck Card — Dependent Claim)

(dependent on 34.1) The system of 34.1 wherein the entrance pathway registry supports configurable journey-fork branching at any point in the enrollment sequence, enabling a single pathway to route prospective members to different cooperative initiative onboarding flows based on member-provided context, while guaranteeing all branches converge to the single canonical cooperative membership class.

Claim 34.5 (Many Doors / Cue Deck Card — Dependent Claim)

(dependent on 34.1) The system of 34.1 wherein the attribution ledger composes with the cooperative's Marks earnings architecture such that inviting members whose referred enrollment occurs within a configurable attribution window

(default: 90 days) receive a permanent attribution-earnings configuration enabling ongoing cooperative Marks earnings attributable to their network development work.

Claim Group 35: Hexadecimal Machine Code — Compact Binary AI-to-AI Wire Format

35.1 (independent) — Hexadecimal Machine Code Compact Dispatch Frame

A system for compact vendor-resilient artificial intelligence substrate worker dispatch, comprising: a persistent AI host application maintaining a registry of active Callable Substrate Workers each identified by a role designator and a connection endpoint; a Hexadecimal Machine Code frame encoder that, upon receiving a task from the host application's orchestration layer, produces a compact binary-encoded dispatch frame comprising: a fixed-width frame header encoding the source host identifier, target substrate worker role designator, task class identifier, and a frame sequence number in hexadecimal notation; a variable-length task payload field encoding task parameters in a typed schema defined by the task class; a ratify-gate field specifying the empirical success criterion the substrate worker must confirm upon task completion; and a frame checksum field; a frame decoder in each registered substrate worker that resolves the frame header to the worker's assigned task handler, extracts typed task parameters from the payload field, and emits a response frame in the same compact encoding; and a human-readable overlay generator that, upon request, renders the frame contents as structured log text for audit purposes without requiring that the transmission format itself be human-readable. (Source: Founder direct BP085 session c9afad17;

canon_hexadecimal_machine_code_mnemosynec_wire_format_consolidation_bp085.eb1et.md)

35.2 (dependent on 35.1) — SSPS Encoding Substrate Composition

The system of Claim 35.1, wherein the Hexadecimal Machine Code frame format serves as the binary encoding substrate for the Stitchpunk Sock Puppet Speak (SSPS) wire-format protocol, such that: SSPS defines the schema of task-dispatch messages exchanged between cooperative substrate workers; and Hexadecimal Machine Code provides the compact binary encoding of those messages, replacing the verbose markdown encoding used in prior SSPS versions; wherein the combination of SSPS schema and Hexadecimal Machine Code encoding produces a dispatch protocol that is simultaneously schema-validated (SSPS layer) and payload-compact (Hexadecimal Machine Code layer), reducing per-dispatch token consumption relative to markdown-formatted equivalents while maintaining full schema compatibility with the SSPS cooperative substrate worker communication standard. (Source: SSPS canon BP055;

canon_sock_puppets_stitchpunks_callable_substrate_workers_bp085.eb1et.md; Founder direct BP085)

35.3 (dependent on 35.1) — Substrate Address Composition via Hexadecimal Coordinate Space

The system of Claim 35.1, wherein the frame header's target substrate worker role designator is resolved through a hexadecimal coordinate address drawn from the Soccerball DAG addressing space, comprising nine hexadecimal digits per coordinate face, such that: each registered substrate worker is assigned a unique Soccerball DAG coordinate as its persistent network address; the dispatch frame header encodes the target address in the same hexadecimal digit vocabulary used throughout the cooperative substrate's peer-to-peer mesh routing layer; and the frame's source identifier similarly encodes the dispatching host's Soccerball DAG address, enabling end-to-end traceability of dispatch frames through the cooperative mesh without requiring a separate address translation layer. (Source: Soccerball DAG recursive substrate addressing BP060;

canon_pocket_universe_soccerball_dag_recursive_substrate_addressing_bp060.eb1et.md; Founder direct BP085)

Claim 35.4 (Hexadecimal Machine Code — Dependent Claim)

(dependent on 35.1) The system of 35.1 wherein the Hexadecimal Machine Code frame includes a priority field in the fixed-width header encoding the task's urgency class drawn from a configurable priority scale (default: 0x00 = lowest through 0xFF = highest), enabling receiving substrate workers to implement priority-queue dispatch for incoming frames.

Claim 35.5 (Hexadecimal Machine Code — Dependent Claim)

(dependent on 35.1) The system of 35.1 wherein the human-readable overlay generator produces frame audit log entries in a structured format compatible with the cooperative's substrate eb1et store, enabling every substrate worker dispatch frame to be archived as a verifiable substrate record with source address, target address, task class, payload hash, ratify-gate specification, and frame checksum.

Claim Group 36: Hot-Swappable Cognitive Core (Brain-Swap) — Vendor-Resilient AI Orchestrator

36.1 (independent) — Method for Hot-Swap of Cognitive Core with State Preservation via Substrate Re-Weave

A method for hot-swapping the cognitive core of a persistent AI orchestrator application with full session-state preservation, comprising: maintaining a substrate state store comprising vendor-agnostic knowledge records (eb1ets) encoding the orchestrator's current session context, domain assignments, task history, and user preferences independently of any AI vendor's conversation history or session state; receiving a cognitive core swap instruction specifying a target AI vendor or model from a Brain Registry; serializing the current session state by executing a substrate re-weave operation that produces a context snapshot from the substrate state store sufficient to rehydrate the incoming

cognitive core to equivalent session awareness; deactivating the outgoing cognitive core's connection; activating an adapter shim for the incoming cognitive core's vendor API; dispatching the serialized context snapshot to the incoming cognitive core via the adapter shim to rehydrate session context; and resuming orchestrator operation with the incoming cognitive core such that the user session continues without interruption, context loss, or re-initialization requirement, wherein the user's accumulated substrate knowledge remains intact because it resides in the substrate state store rather than in any vendor's conversation history. (Source: Founder direct BP085;

canon_mnemo_brain_swap_pluggable_cognitive_core_hot_swappable_bp085.eblet.md; composes with Substrate Theorem BP061)

36.2 (dependent on 36.1) — Cognitive Core Interface Contract Specification

The method of Claim 36.1, wherein the Brain Registry qualifies cognitive core candidates against a Cognitive Core Interface (CCI) contract comprising: an input wire format requirement specifying that the cognitive core must accept structured task dispatch frames in the compact encoding defined by the Hexadecimal Machine Code wire format of Claim Group 35, or an equivalent vendor-agnostic structured format; an output wire format requirement specifying that the cognitive core must return structured response frames with ratify-gate confirmation fields; a tool-calling capability requirement specifying that the cognitive core must support the cooperative's MCP tool-calling standard enabling the cognitive core to invoke registered substrate tools; a declared minimum context window sufficient to receive and process a full substrate re-weave context snapshot without truncation; and a swap eligibility declaration indicating whether the cognitive core supports hot-swap mid-session participation; wherein only cognitive cores satisfying all CCI contract requirements are eligible for inclusion in the Brain Registry and selection for cognitive core dispatch, such that the CCI contract enforces vendor-agnostic interoperability at the orchestrator's cognitive layer. (Source:

canon_mnemo_brain_swap_pluggable_cognitive_core_hot_swappable_bp085.eblet.md; composes with

canon_hexadecimal_machine_code_mnemosynec_wire_format_consolidation_bp085.eblet.md CG35)

36.3 (dependent on 36.1) — Per-Task and Per-Session Swap Policy with Cost Routing per Consult-Don't-Rent

The method of Claim 36.1, wherein cognitive core swap instructions are generated by one of: (a) a per-task swap policy that classifies each incoming task against a task-class routing table and selects the minimum-capable cognitive core tier sufficient for the task class, consistent with the Consult-Don't-Rent routing discipline of Claim 33.3, restoring the prior cognitive core upon task completion; (b) a per-session swap policy declared at session open by the user or by a rotation policy defined in the orchestrator's configuration, wherein the selected cognitive

core persists for the session duration; or (c) a direct user instruction issued mid-session naming a specific cognitive core or vendor; and wherein each cognitive core swap event generates a cost-delta disclosure record stating the incoming cognitive core's per-task cost, the delta relative to the prior cognitive core's cost for equivalent tasks, and the user-visible rationale for the swap, composing with the Unseen Tax anti-pattern disclosure discipline of Claim Group 30, such that the user maintains full visibility into per-swap cost implications without requiring that such visibility be sought in application settings. (Source:

canon_mnemo_brain_swap_pluggable_cognitive_core_hot_swappable_bp085.eblet.md; composes with

canon_consult_ai_flagships_dont_rent_the_c40_aircraft_bp085.eblet.md CG33.3; composes with Unseen Tax CG30)

Claim 36.4 (Brain-Swap Hot-Swap Cognitive Core — Dependent Claim)

(dependent on 36.1) The method of 36.1 wherein the Brain Registry catalogs adapter shim implementations for: Anthropic Claude API; OpenAI GPT API; Google Gemini API; Meta Llama local inference; Mistral local inference; and Ollama-managed local model instances; wherein each adapter shim normalizes the vendor-specific API format to the Cognitive Core Interface (CCI) contract such that vendor API changes require only shim updates.

Claim 36.5 (Brain-Swap Hot-Swap Cognitive Core — Dependent Claim)

(dependent on 36.1) The method of 36.1 wherein the per-task swap policy composes with the MedLab Reins Assignment mechanism (Claim Group 32) such that the domain assignment registry governs which cognitive core is assigned per-domain, enabling domain-specialist routing at the cognitive core layer, with return to the session's primary cognitive core upon task completion.

Claim Group 37: Mimic Trunks — Gate-and-Tunnel Partner Cooperative Volume Benefits

Background and Problem Statement

Cooperative platform pages benefit from external design expertise. Prior architectures provide no mechanism enabling a partner to publish a parallel redesigned variant (Mimic Trunk), route users through the partner's path while requiring the cooperative's own authentication gate, attribute value to the partner based on user preference, and automatically extend cooperative-tier volume pricing benefits to the partner.

The Founder's formulation: "BIGGEST is to get someone to redesign the pages (on mimic trunks) so that it all works better. And if they do that, and people follow that path, or like it, then they totally get credit — but it has to merge with ours, and then they get their portion of people using their tunnel after entering at

our gate. This is a common theme for all businesses that link up with LB — so that they can take advantage of our Volume Pricing and Member Benefits like Stripe Card Processing and AI pools.” (Founder direct, BP084, 2026-06-15.)

Detailed Description

Claim 37.1 (Mimic Trunks Gate-and-Tunnel — Independent System Claim)

A computer-implemented cooperative partner page-variant architecture comprising: - (a) a Mimic Trunk registry storing partner-published page variant records, each comprising a partner identifier, a canonical cooperative page identifier, a public variant URL, a fork timestamp, and a trust-score record initialized at zero; - (b) a cooperative gate mechanism requiring all user sessions initiating through any registered Mimic Trunk variant to authenticate through the cooperative’s Thorax heartbeat endpoint before proceeding to the partner’s variant flow, wherein Thorax authentication is non-delegatable to the partner and cannot be replaced by partner-controlled authentication; - (c) a tunnel routing mechanism that, after successful gate authentication, presents the member with the partner’s variant flow and records the tunnel entry event in a usage ledger attributed to the partner identifier; - (d) a merge evaluation pathway enabling a partner to submit their Mimic Trunk variant for A/B empirical receipt evaluation against the canonical page, wherein canonical page content is updated to incorporate improvements verified as superior per a configurable improvement metric, and the contributing partner is recorded in the canonical page’s contributor manifest upon merge; - (e) a tunnel revenue share mechanism computing a partner’s periodic earnings as a function of the count of members who entered through the partner’s Mimic Trunk variant, the cooperative-class value realized per entering member, and a configurable partner share percentage; wherein revenue share is denominated in cooperative Marks or processed through the Fiat Pipeline Rail per Claim Group 12; and - (f) a contributor attribution record written to the cooperative IP ledger at merge time comprising the partner identifier, canonical page identifier, merge timestamp, and a permanent attribution flag, wherein attribution persists in the IP ledger regardless of whether the partner continues to maintain their Mimic Trunk after merge. **37.2** (dependent on 37.1) The system of 37.1 wherein the cooperative gate mechanism enforces cooperative authentication sovereignty such that a partner who operates a Mimic Trunk variant cannot create cooperative member sessions independently of the Thorax heartbeat endpoint, issue cooperative credentials or Marks, intercept or modify Thorax authentication payloads, or operate the variant in a mode that bypasses Thorax authentication for any subset of users; ensuring that cooperative member identity and credential issuance remain structurally controlled by the cooperative regardless of the number or variety of registered variants. **37.3** (dependent on 37.1) The system of 37.1 wherein the tunnel revenue share mechanism distributes tunnel-attributed value among: a partner share allocated to the Mimic Trunk operator; a cooperative operating share; a maker credit pool

allocated proportionally to cooperative members who contributed to the canonical page being forked; and a platform engineering reserve; wherein the sum of all portions equals total tunnel-attributed value for the period, and no fiat proceeds are held by the cooperative beyond the transaction settlement window per the Substitution Rail fiat-pipeline prohibition of Claim Group 12. **37.4** (dependent on 37.1) The system of 37.1 wherein a partner whose Mimic Trunk variant is registered is automatically enrolled in the cooperative's volume pricing extension, receiving cooperative-tier pricing for: merchant payment processing through the cooperative's negotiated processing agreements; artificial intelligence inference access through the cooperative's collective AI pool agreements; and cooperative compute access through the Federation Node Frontier capsule rental mechanism (Claim Group 9) at member rates; wherein volume pricing benefits activate upon first registration and remain active while the registration is in good standing.

Empirical Receipt

Founder direct BP084 (2026-06-15): "Biggest Bounty Poster, that is." Canon eblet canon-mimic-trunks-gate-and-tunnel-partner-cooperative-volume-benefits-bp084 ratified. Canonical vocabulary (Mimic Trunk/Gate/Tunnel/Merge) ratified. Volume pricing package (Stripe Card Processing, AI pools, Federation Node Frontier compute) ratified.

Claim 37.5 (Mimic Trunks Gate-and-Tunnel — Dependent Claim)

(dependent on 37.1) The system of 37.1 wherein Mimic Trunks volume aggregation credits are time-stamped and sorted by cooperative enrollment date such that earlier-enrolled members receive priority in tunnel allocation when total demand exceeds contracted partner capacity.

Claims

The detailed description above presents the claims framework for this provisional patent application. Formal independent and dependent claims will be crystallized in the corresponding non-provisional application. The applicant reserves the right to file a non-provisional application claiming priority to this provisional application for all innovations disclosed herein.

The innovations disclosed in Claim Groups 1-37 collectively constitute a cooperative-architecture local-first AI substrate platform composing the following independent and dependent claim families in the non-provisional application:

- Independent Claims: Canonical Plow Pipeline (CG1), Andon Cord Self-Policing (CG3), BMV Scoring (CG4), Per-Domain Isolation (CG5), MEMORY.md Auto-Loading (CG6), Six-Folder Layout (CG7), MIC Conductor Orchestration (CG16), Federated Andon 3-Tier Escalation (CG17), The Diagnosis Broadcast (CG18), Salt-Level Persistence Selector (CG19), Glow Mechanic (CG20), Three-Salt-Layer Architecture (CG21), Hardware-

Tier Model Selection (CG22), Plow Resume from Checkpoint (CG23), Onboarding Lifecycle Auto-Advance (CG24), Constellation Switchboard Shape Mapping (CG25), Substitution Rail Cooperative Marketplace (CG26).

- Claim Groups 31-37
(MnemosyneC/MedLab/NetLinkWebNode/ManyDoors/HexMachineCode/Bra inSwap/MimicTrunks): Independent Claims: CG31.1, CG32.1, CG32.3, CG33.1, CG33.3, CG34.1, CG35.1, CG36.1, CG37.1. Dependent claims: see respective claim group detailed descriptions.
 - Innovation Area 35 (Truth Integrity Chain): Independent Claims CG28.1, CG28.4, CG28.5; Dependent Claims CG28.2, CG28.3
 - Innovation Area 36 (Code Breakers Guild): Independent Claims CG29.1, CG29.3; Dependent Claims CG29.2, CG29.4, CG29.5
 - Innovation Area 37 (Unseen Tax): Independent Claims CG30.1, CG30.2, CG30.3; Dependent Claims CG30.4, CG30.5
 - Claim Group 27 (Tab UX + Test-Net by Design): Independent Claims CG27.1, CG27.4; Dependent Claims CG27.2, CG27.3, CG27.5, CG27.6
- v03–v05 additional claim families (CG28–37): independent claims per CG detailed in respective descriptions above; dependent claims meet ≥ 4 -per-independent requirement throughout. ode Frontier (CG9, dependent on CG7/CG12), MemoryWallPyramid (CG10, dependent on CG7), Package Store Confectionary (CG11, dependent on CG12), Substitution Rail Triple-Mode Exchange (CG12, independent and compositional), GPQA Diamond Benchmark (CG13, dependent on CG1), EAC Pheromone Decay (CG14, dependent on CG7), MMLU-Pro Benchmark (CG15, dependent on CG1/CG5).

Brief Description of Drawings

Drawings to be submitted with non-provisional application:

- Figure 1: Canonical Plow Pipeline Architecture (nine-stage sequential flow)
- Figure 2: Andon Cord Re-Plow Loop (state machine with three-retry cap)
- Figure 3: Per-Domain Isolation Architecture (fourteen parallel tracks)
- Figure 4: Six-Folder Substrate Layout (directory hierarchy)
- Figure 5: MemoryWallPyramid Addressing (soccerball-DAG recursive structure)
- Figure 6: Package Store Confectionary (three-stall marketplace)
- Figure 7: Substitution Rail Triple-Mode Exchange
- Figure 8: MEMORY.md Self-Context Auto-Loading cycle
- Figure 9: MIC Conductor-Class Orchestration (fleet discovery, greedy partition, parallel dispatch, failure recovery)
- Figure 10: Federated Andon 3-Tier Escalation (Tier 1 widened roster → Tier 2 constellation → Tier 3 Diagnosis state machine)
- Figure 11: The Diagnosis Persistent Question Broadcast (post lifecycle, 3-voter concordance, Marks bounty, substrate write-back)

- Figure 12: Salt-Level Persistence-Tier Selector (three-tier segmented control with auto-suggest heuristics)
- Figure 13: Glow Mechanic Five-Tier Scale (0 / 1-9 / 10-99 / 100-999 / 1000+ Marks visual intensity)
- Figure 14: Three-Salt-Layer Architecture (L1 Substrate Salt → L2 Federation Salt → L3 Human Salt escalation flow)
- Figure 15: Hardware-Tier Model Selection (RAM detection → tier classification → model pull flow)
- Figure 16: Plow Resume from Checkpoint (checkpoint write cycle, interrupt detection, resume decision, cleanup)
- Figure 17: Onboarding Lifecycle Auto-Advance (Stage A/B/C state machine, stuck-stage detector, reset-diagnostic, auto-advance polling)
- Figure 18: Constellation Switchboard NotCents Shape Mapping (three-shape four-state matrix with Conductor crown overlay)
- Figure 19: Substitution Rail Cooperative Marketplace (three-rail payment dispatch, XP/RP, Medallion provenance stamp, creator distribution)
- Figure 20: Five-Field Truth Integrity Record (block diagram)
- Figure 21: Consequence Trace Loop 10 example
- Figure 22: Dependency Propagation Loop 12 example
- Figure 23: Hasty-Generalization Guard flowchart
- Figure 24: Four-Tier Knowledge Claim Progression (UNTESTED to IMMUTABLE)
- Figure 25: Code Breaker Member Tier Progression
- Figure 26: Negative-Knowledge Tokens Dashboard
- Figure 27: Per-Call Cost Surface Interaction Flow
- Figure 28: Cross-Vendor Aggregate Dashboard via Personal Subscription Steward
- Figure 29: Model Selection Decision Flowchart
- Figure 30: MnemosyneC Persistent Host Architecture
- Figure 31: Session-Boundary Re-Weave and Continuity-Lift cycle
- Figure 32: MedLab Concoction Lifecycle
- Figure 33: Reins Assignment Registry and handoff ceremony
- Figure 34: NetLinkWebNode Peer Borrowing Architecture
- Figure 35: MIC Election Algorithm
- Figure 36: Many Doors One Cooperative Entrance Pathway Registry
- Figure 37: Cue Deck Card System
- Figure 38: Hexadecimal Machine Code Frame Structure
- Figure 39: SSPS and Hex Machine Code Composition
- Figure 40: Brain Registry and Cognitive Core Interface Contract
- Figure 41: Substrate Re-Weave State Preservation
- Figure 42: Mimic Trunk Registry and Gate-and-Tunnel Flow
- Figure 43: Tunnel Revenue Share Distribution
- Figure 44: Volume Pricing Extension Architecture

- Figure 20: Five-Field Truth Integrity Record (KNOWN/THEORIES_OPEN/ELIMINATED/DEPENDENCIES_UPSTREAM/APPLICATIONS_DOWNSTREAM block diagram)
 - Figure 21: Consequence Trace Loop 10 example (MOND theory with 3-deep consequence trail)
 - Figure 22: Dependency Propagation Loop 12 example (KNOWN K-001 update propagates through 5+ downstream records)
 - Figure 23: Hasty-Generalization Guard (universal claim to sample-size check to counter-evidence search)
 - Figure 24: Four-Tier Knowledge Claim Progression (UNTESTED to TESTED to FORGED to GOLD_REFINED_BY_FIRE with reversal path)
 - Figure 25: Code Breaker Member Tier Progression (Initiate to Journeyman to Master to Refiner of Gold)
 - Figure 26: Negative-Knowledge Tokens Dashboard (per-member ratio; aggregate cooperative dashboard)
 - Figure 27: Per-Call Cost Surface Interaction Flow
 - Figure 28: Cross-Vendor Aggregate Dashboard via Personal Subscription Steward
 - Figure 29: Model Selection Decision Flowchart (cheapest-capable to demonstrated insufficiency to escalation)
 - Figure 30: MnemosyneC Persistent Host Architecture (local app to substrate store to vendor routing)
 - Figure 31: Session-Boundary Re-Weave and Continuity-Lift cycle
 - Figure 32: MedLab Concoction Lifecycle (trigger to ingredient manifest to execution to ratify gate)
 - Figure 33: Reins Assignment Registry and handoff ceremony
 - Figure 34: NetLinkWebNode Peer Borrowing (Constellation Switchboard to lending consent to Marks settlement)
 - Figure 35: MIC Election Algorithm (reputation-weighted; coordinator routing; re-election on departure)
 - Figure 36: Many Doors One Cooperative Entrance Pathway Registry
 - Figure 37: Cue Deck Card System (recipient context to value proposition to journey fork to attribution)
 - Figure 38: Hexadecimal Machine Code Frame Structure
 - Figure 39: SSPS + Hex Machine Code Composition
 - Figure 40: Brain Registry and Cognitive Core Interface Contract
 - Figure 41: Substrate Re-Weave State Preservation method
 - Figure 42: Mimic Trunk Registry and Gate-and-Tunnel Flow
 - Figure 43: Tunnel Revenue Share Distribution
 - Figure 44: Volume Pricing Extension Architecture
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Abstract

A cooperative-architecture AI system for local-first knowledge substrate growth on consumer hardware. Nine-stage canonical plow pipeline (Spider, Sprite, FireGuard staggered swarm, Miner filter, Saladin adversarial fence, Furnace, Three Fates concordance, BMV Scribe, Detective/Andon) across fourteen isolated domains. MIC Conductor fleet orchestration; Federated Andon 3-Tier Never-Give-Up escalation; Diagnosis persistent human broadcast; Salt-Level persistence selector; Glow Mechanic attention surfacing; Hardware-Tier model selection; Plow Resume from Checkpoint; Onboarding Auto-Advance; Constellation Switchboard visual primitive; Substitution Rail triple-mode marketplace (83.3% creator distribution); Tab UX toggle; Test-Net by Design; Truth Integrity Chain five-field epistemic record with Loops 10/11/12; Code Breakers Guild Gold Refined by Fire; Unseen Tax real-time AI cost transparency; MnemosyneC-as-Interface persistent host; MedLab Concoctions; NetLinkWebNode peer borrowing; Many Doors unified membership; Hexadecimal Machine Code wire format; Brain-Swap hot-swappable core; Mimic Trunks gate-and-tunnel partner architecture. Empirical receipt: 97.1% (68/70 MMLU-Pro), consumer hardware, no paid API keys. Pledge #2260: defensive patent use only.

Filing Gate Status

PUBLICATION GATE: HARD CLOSED

Per Fire Control Directive (#2330, Prov 22): this specification is internal-only pending Founder review and explicit USPTO Patent Center submission authorization.

Knight scope: build and stage this specification. COMPLETE. Founder scope: review this document; upload to USPTO Patent Center; authorize submission.

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Mechanical gate: `USPTO_SUBMISSION_AUTHORIZED=false` (this specification)

Liana Banyan Corporation · Inventor: J. Jones · Provisional Patent Application 22 BP083–BP086 Plow Cycle · Sonnet 4.6 SEG · June 2026 37 Claim Groups · 44 Innovation Areas · Incremental over Provisional Applications 1–21 v06_AA_STYLE: A&A canonical reformat — shorter title + TOC + heading hierarchy + claim notation